

Chapter 1

Introduction

- Data Communication
- Networks
- Protocols and Standards
- Standard Organizations



Data and Information

- **Data** refers to the raw facts that are collected while **information** refers to processed data that enables us to take decisions.
- Ex. When result of a particular test is declared it contains data of all students, when you find the marks you have scored you have the information that lets you know whether you have passed or failed.
- The word **data** refers to any information which is presented in a form that is agreed and accepted upon by its creators and users.

Data Communication

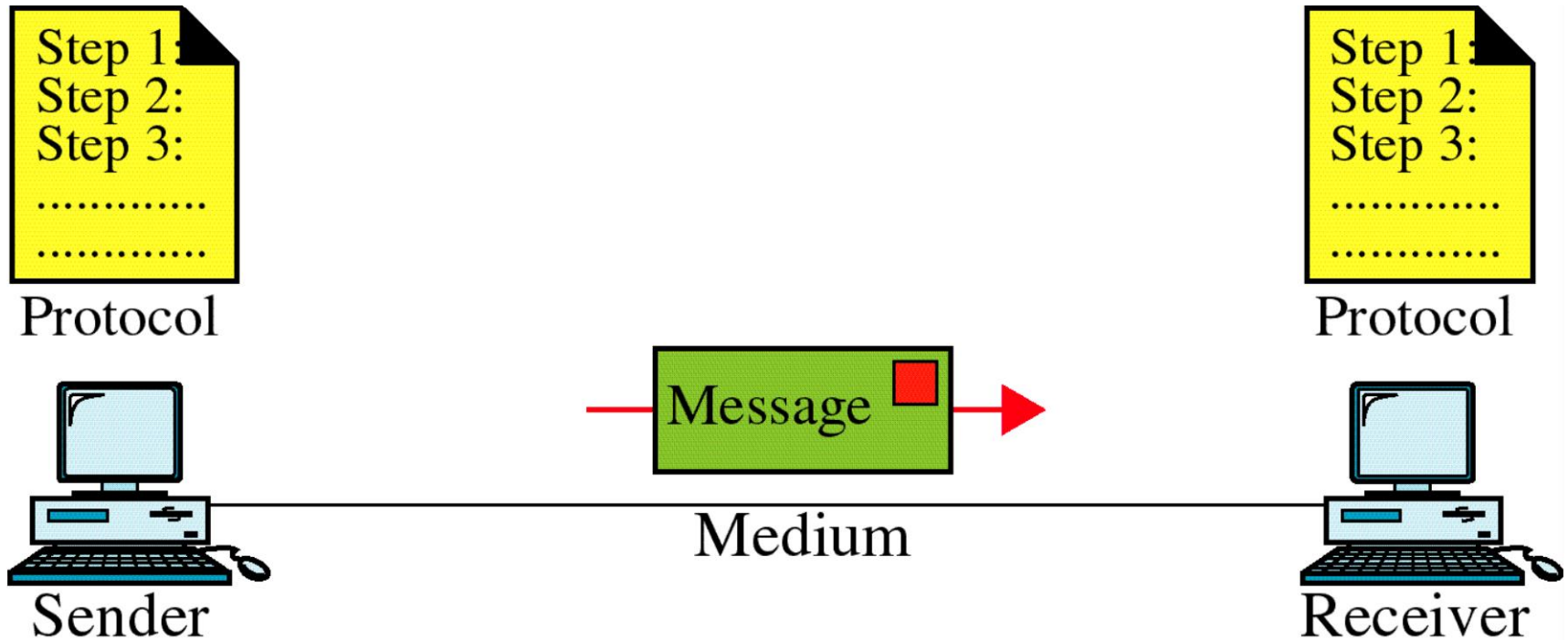
- Data Communication is a process of exchanging data or information
- In case of computer networks this exchange is done between two devices over a transmission medium.
- This process involves a communication system which is made up of hardware and software. The hardware part involves the sender and receiver devices and the intermediate devices through which the data passes. The software part involves certain rules which specify what is to be communicated, how it is to be communicated and when. It is also called as a Protocol.

Characteristics of Data Communication

The effectiveness of any data communications system depends upon the following four fundamental characteristics:

- 1. **Delivery:** The data should be delivered to the correct destination and correct user.
- 2. **Accuracy:** The communication system should deliver the data accurately, without introducing any errors. The data may get corrupted during transmission affecting the accuracy of the delivered data.
- 3. **Timeliness:** Audio and Video data has to be delivered in a timely manner without any delay; such a data delivery is called real time transmission of data.
- 4. **Jitter:** It is the variation in the packet arrival time. Uneven Jitter may affect the timeliness of data being transmitted.

Data Communication System Components



Components of a Data Communication System

- **1. Message:** Message is the information to be communicated by the sender to the receiver.
- **2. Sender:** The sender is any device that is capable of sending the data (message).
- **3. Receiver:** The receiver is a device that the sender wants to communicate the data (message).
- **4. Transmission Medium:** It is the path by which the message travels from sender to receiver. It can be wired or wireless and many subtypes in both.
- **5. Protocol:**
 - It is an agreed upon set or rules used by the sender and receiver to communicate data.
 - A protocol is a set of rules that governs data communication.
 - A Protocol is a necessity in data communications without which the communicating entities are like two persons trying to talk to each other in a different language without know the other language.

Data communication
network criteria

```
graph TD; A[Data communication network criteria] --> B[Performance]; A --> C[Reliability]; A --> D[Security];
```

Performance

Reliability

Security

Basic Concepts

- **Line Configuration**
- **Topology**
- **Transmission Mode**
- **Categories of Networks**
- **Internetworks**

Line configuration

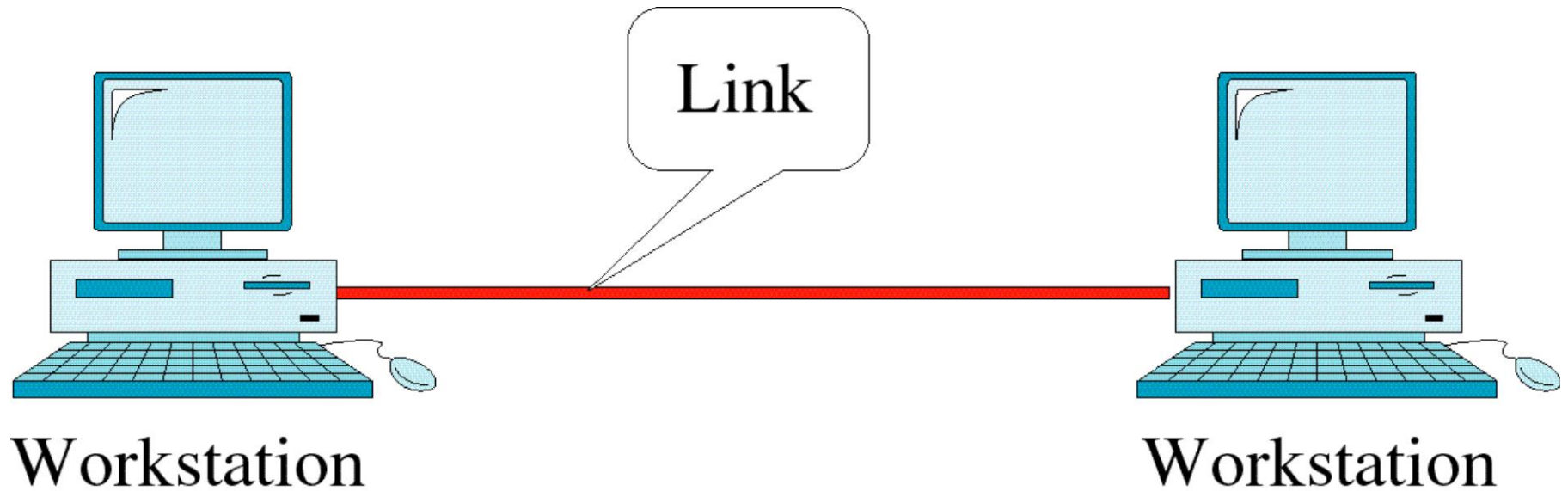
```
graph TD; A[Line configuration] --> B[Point-to-point]; A --> C[Multipoint]
```

A hierarchical diagram with 'Line configuration' at the top. A vertical line descends from it and connects to a horizontal line. From this horizontal line, two vertical lines descend to 'Point-to-point' on the left and 'Multipoint' on the right. All three text boxes have a thick black L-shaped shadow on their bottom-right corners.

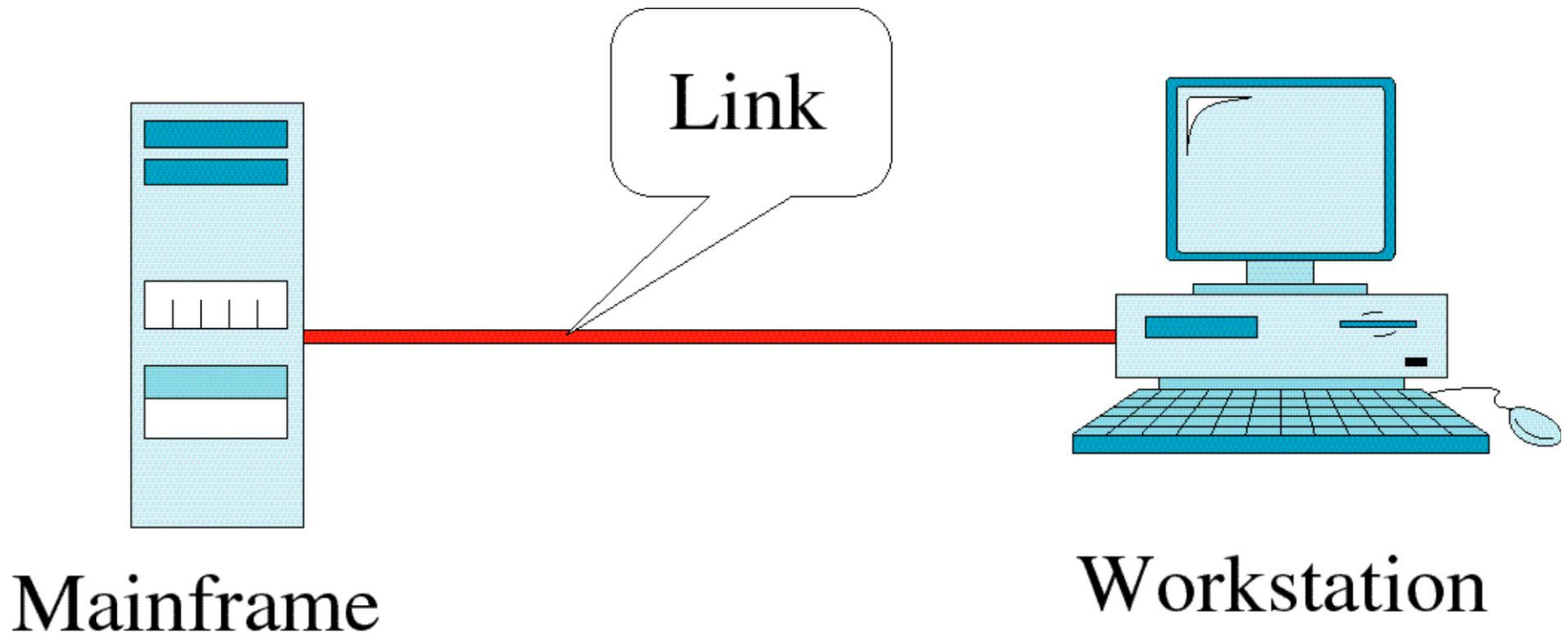
Point-to-point

Multipoint

Point-to-Point Line Configuration



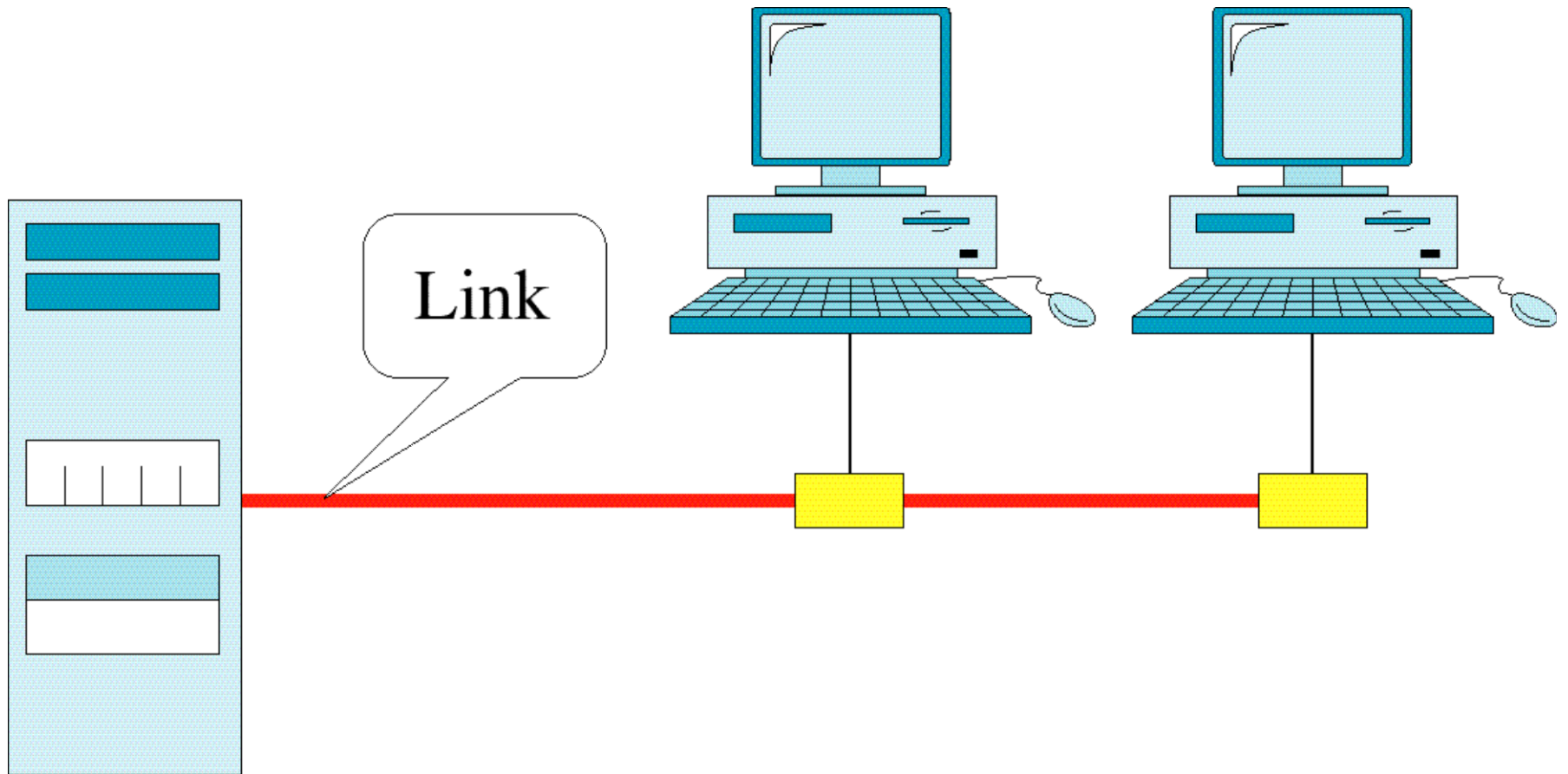
Point-to-Point Line Configuration

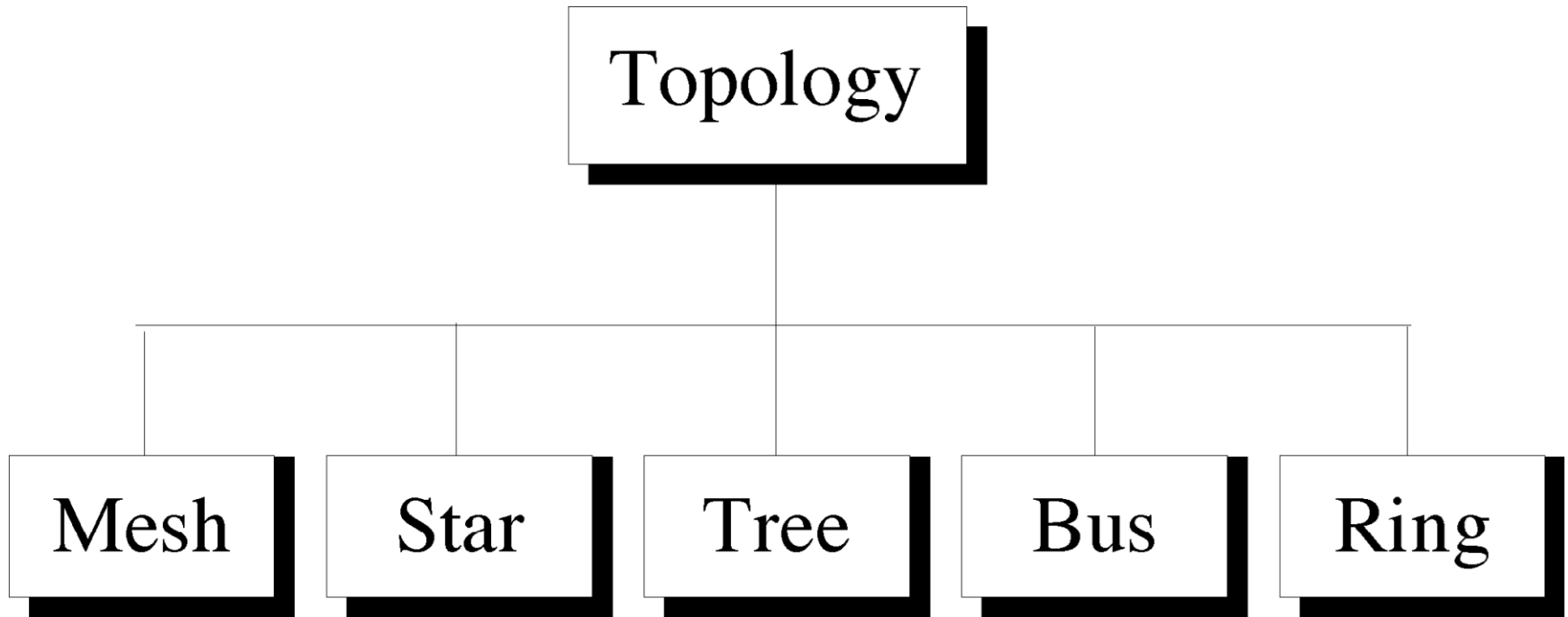


Point-to-Point Line Configuration



Multipoint Line Configuration



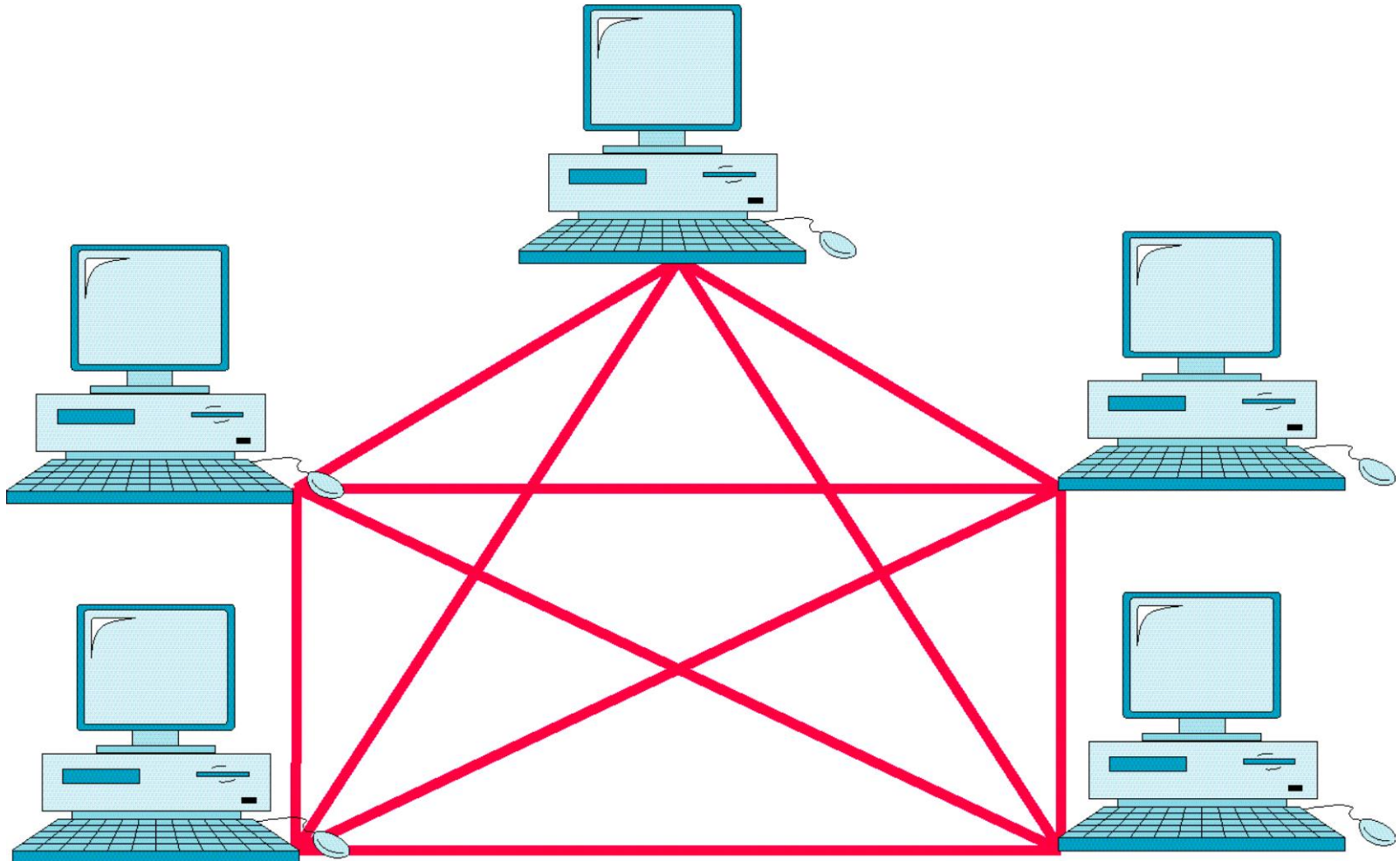


Topology

- Physical and logical network layout
 - – Physical – actual layout of the computer cables and other network devices
 - – Logical – the way in which the network appears to the devices that use it.

Common topologies: – Bus, ring, star, mesh and wireless

Mesh Topology

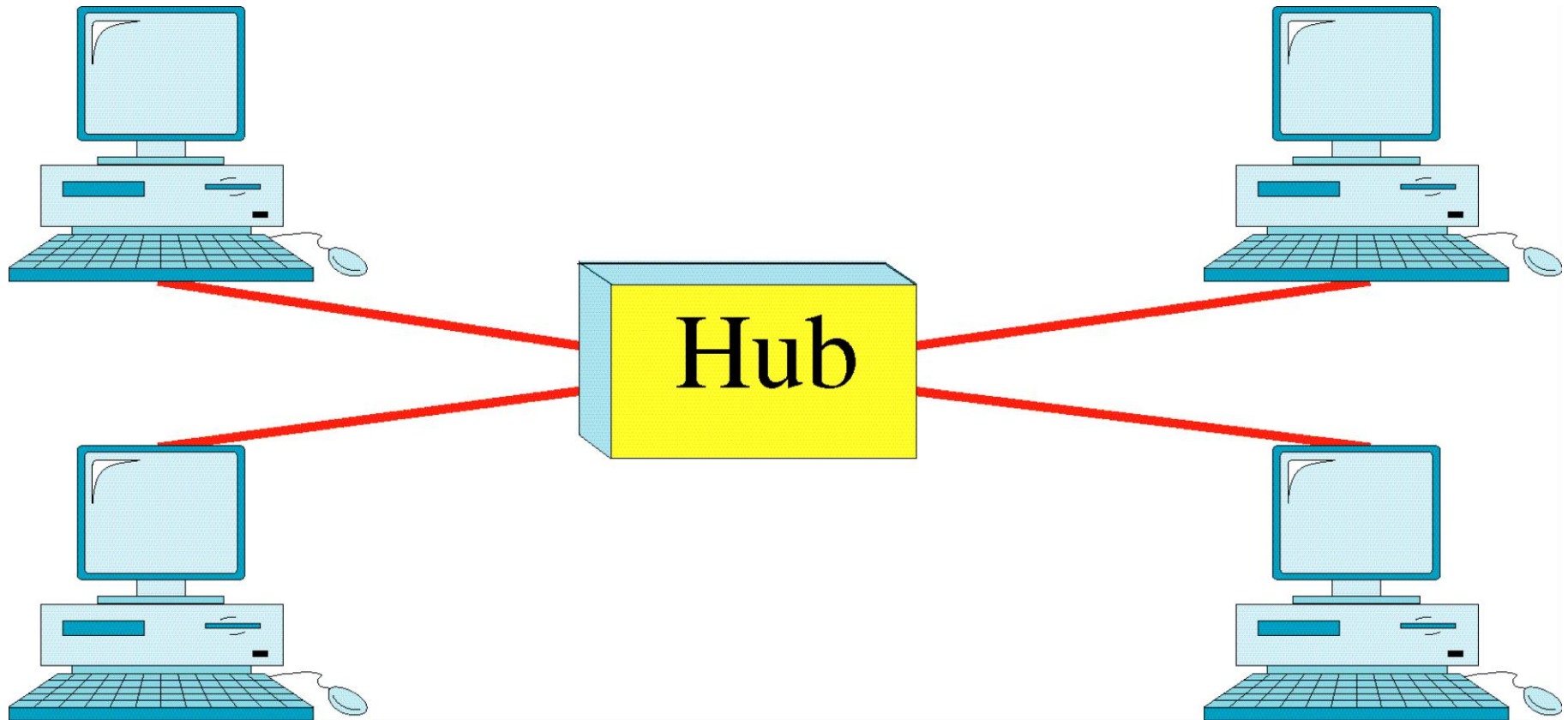


Mesh Topology

- Each computer connects to every other.
- High level of redundancy.
- Rarely used.
 - Wiring is very complicated
 - Cabling cost is high
 - Troubleshooting a failed cable is tricky
 - A variation hybrid mesh
 - create point to point connection between specific network devices, often seen in WAN implementation.

Advantages	Disadvantages
Provides redundant paths between devices	Requires more cable than the other LAN topologies
The network can be expanded without disruption to current uses	Complicated implementation

Star Topology

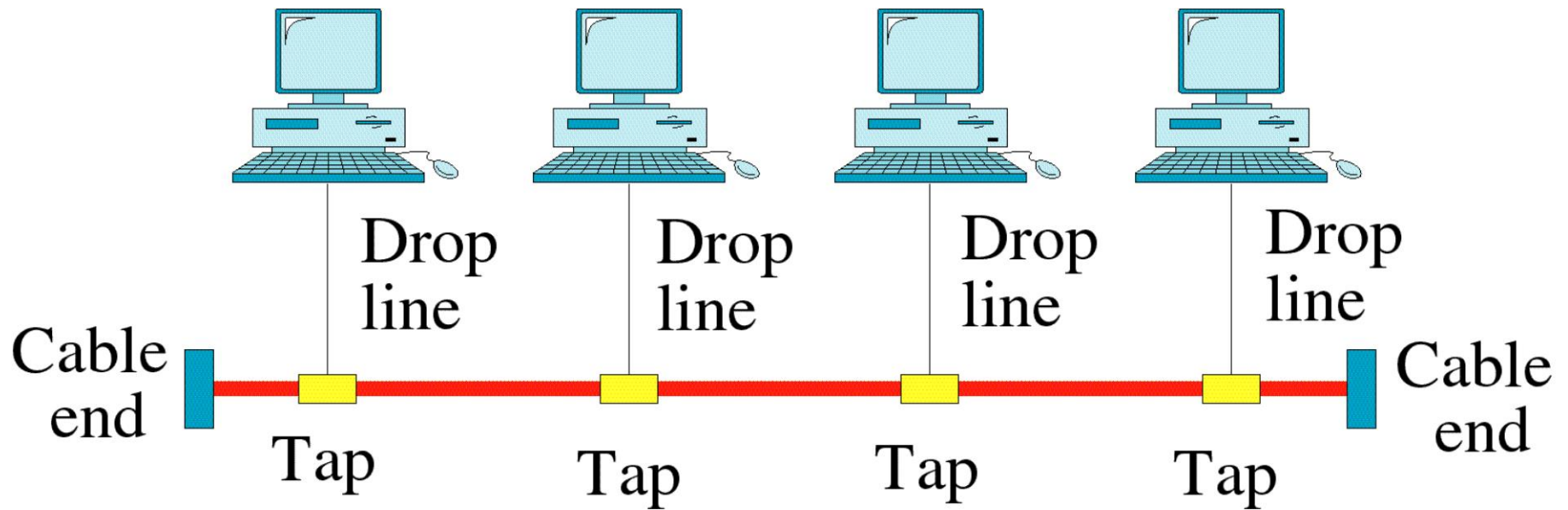


Star Topology

- All computers/devices connect to a central device called hub or switch.
- Each device requires a single cable
- point-to-point connection between the device and hub.
- Most widely implemented
- Hub is the single point of failure

Advantages	Disadvantages
Easily expanded without disruption to the network	Requires more cable
Cable failure affects only a single user	A central connecting device allows for a single point of failure
Easy to troubleshoot and isolate problems	More difficult to implement

Bus Topology

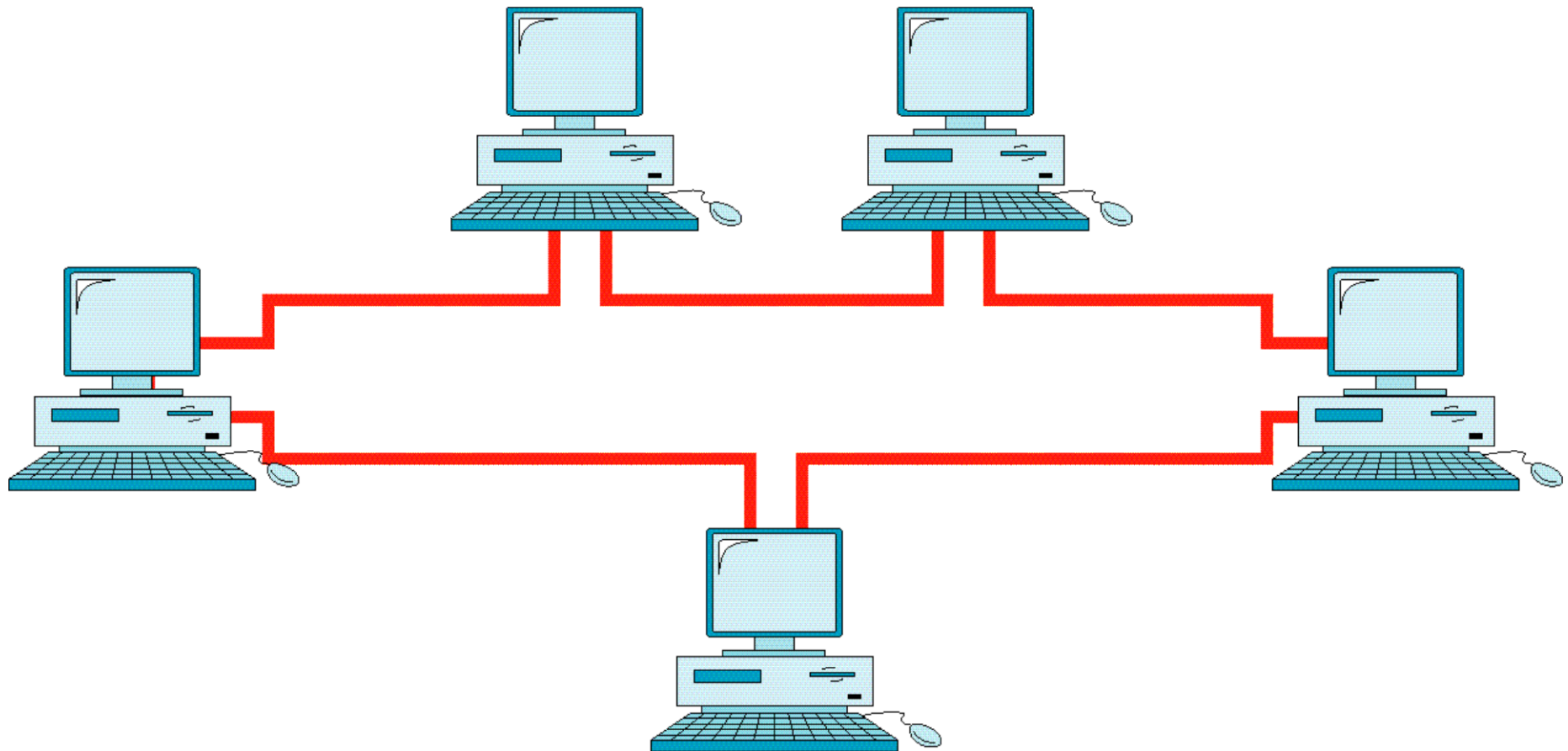


Bus Topology

- Uses a trunk or backbone to which all of the computers on the network connect.
- Systems connect to this backbone using T connectors or taps.
- Coaxial cablings (10Base-2, 10Base5) were popular options years ago.

Advantages	Disadvantages
Cheap and easy to implement	Network disruption when computers are added or removed
Require less cable	A break in the cable will prevent all systems from accessing the network.
Does not use any specialized network equipment.	Difficult to troubleshoot.

Ring Topology



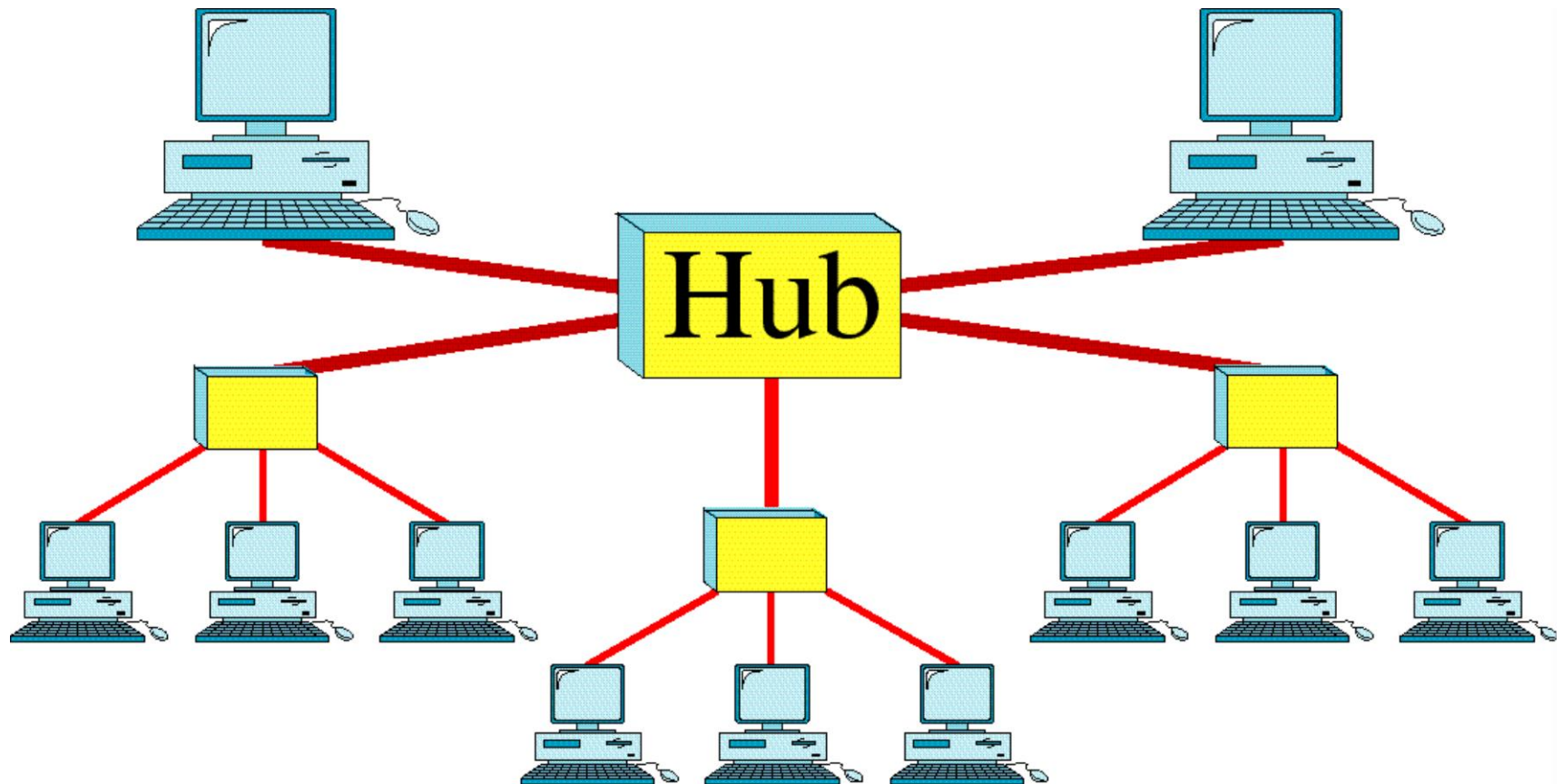
Ring Topology

Logical ring

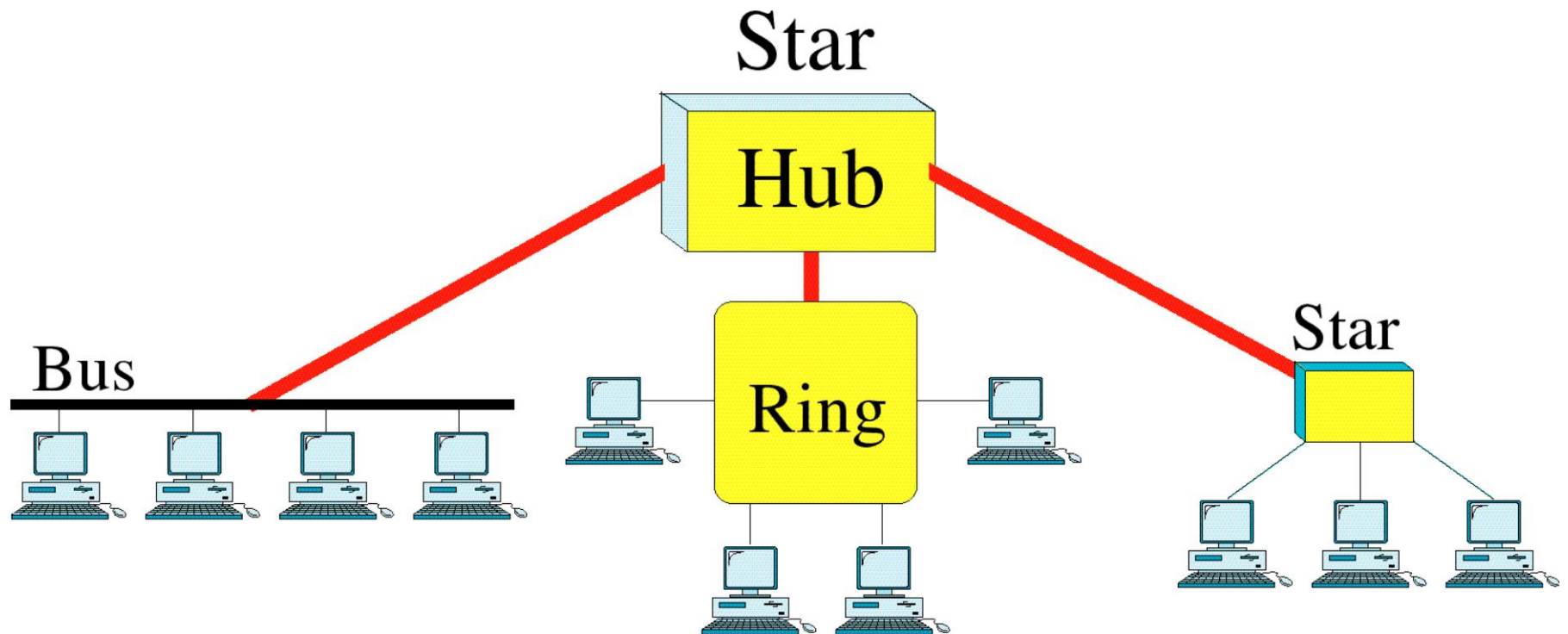
- Meaning that data travels in circular fashion from one computer to another on the network.
- Typically FDDI, SONET or Token Ring technology are used to implement a ring network
- Ring networks are most commonly wired in a star configuration
 - Token Ring has Multi-Station Access Unit (MSAU), equivalent to hub or switch. MSAU performs the token circulation internally.

Advantages	Disadvantages
Cable faults are easily located, making troubleshooting easier	Expansion to the network can cause network disruption
Ring networks are moderately easy to install	A single break in the cable can disrupt the entire network.

Tree Topology



Hybrid Topology



Transmission mode

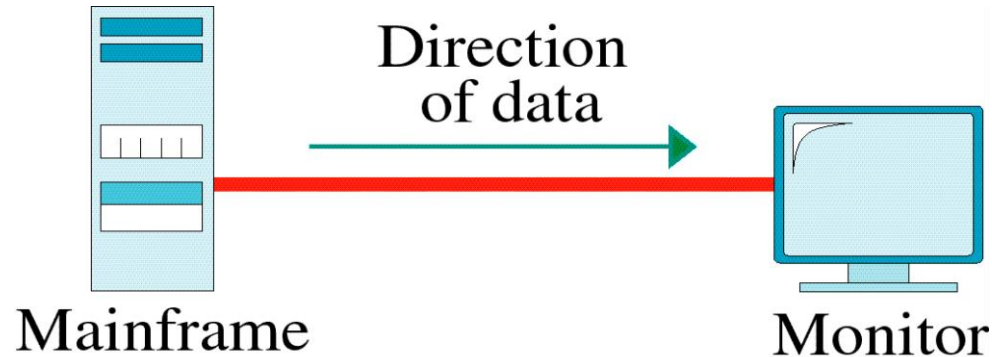
```
graph TD; A[Transmission mode] --> B[Simplex]; A --> C[Half-duplex]; A --> D[Full-duplex]
```

Simplex

Half-duplex

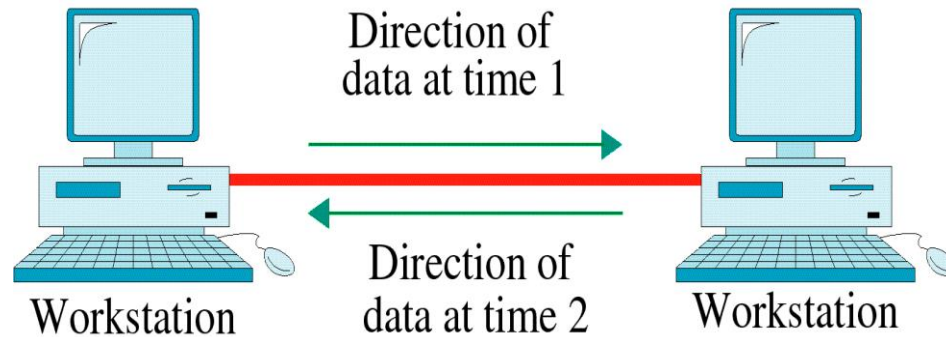
Full-duplex

Simplex



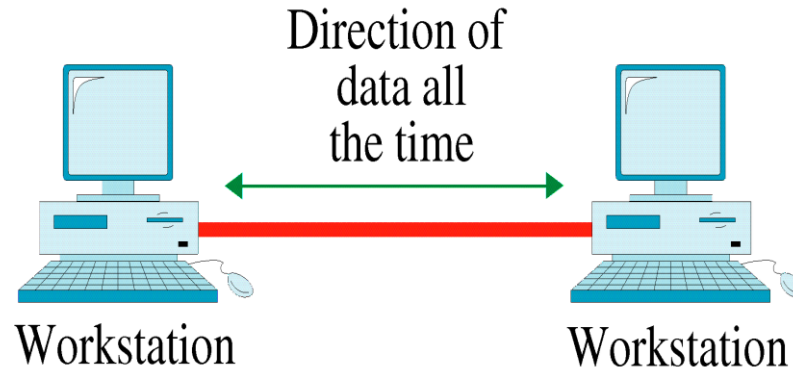
- In Simplex, communication is unidirectional
- Only one of the devices sends the data and the other one only receives the data.
- Example: in the above diagram: a CPU send data while a Monitor only receives data.

Half-Duplex



- In half duplex both the stations can transmit as well as receive but not at the same time.
- When one device is sending other can only receive and vice-versa (as shown in figure above.)
- Example: A walkie-talkie.

Full-Duplex



- In Full duplex mode, both stations can transmit and receive at the same time.
- Example: mobile phones

Computer Networks

Computer Networks are used for data communications

Definition:

A computer network can be defined as a collection of nodes. A node can be any device capable of transmitting or receiving data. The communicating nodes have to be connected by communication links.

A Compute network should ensure

- **reliability** of the data communication process,
- **security** of the data
- **performance** by achieving higher throughput and smaller delay times

Network

```
graph TD; Network[Network] --- LAN[Local area network (LAN)]; Network --- MAN[Metropolitan area network (MAN)]; Network --- WAN[Wide area network (WAN)];
```

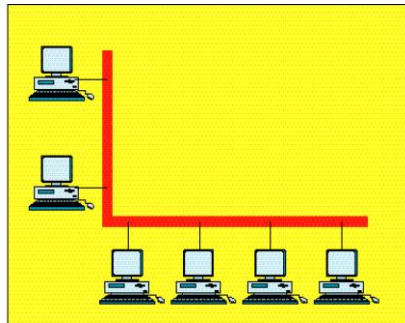
Local area
network
(LAN)

Metropolitan area
network
(MAN)

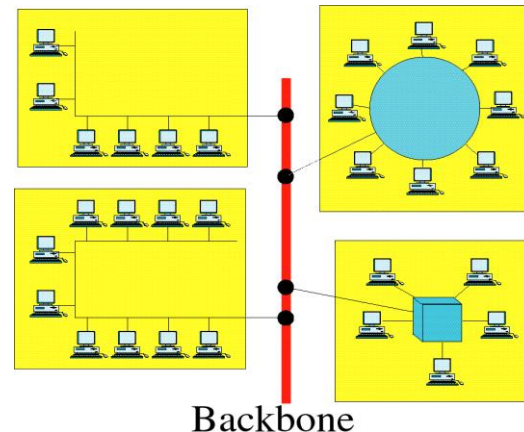
Wide area
network
(WAN)

Local Area Network

Local Area Networks (LAN) is usually limited to a few kilometers of area. It may be privately owned and could be a network inside an office on one of the floor of a building or a LAN could be a network consisting of the computers in a entire building.



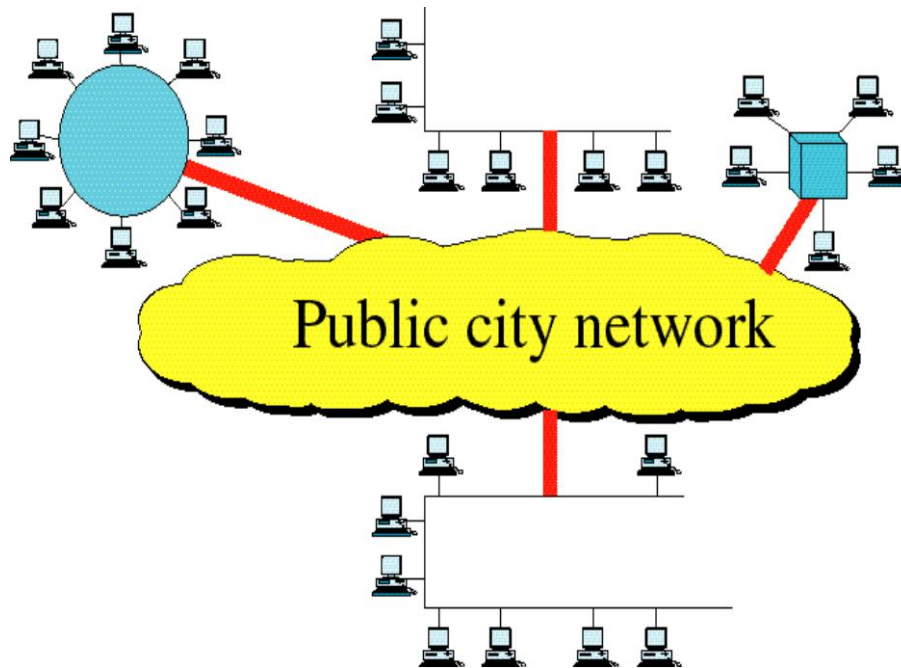
Single building LAN



Multiple building LAN

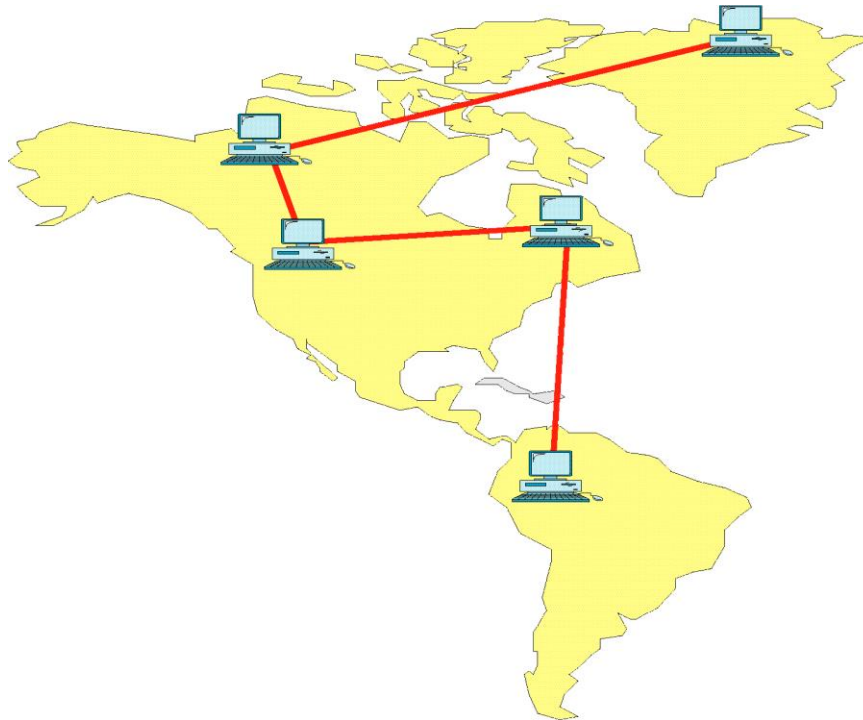
Metropolitan Area Network

Metropolitan Area Network (MAN) is of size between LAN & WAN. It is larger than LAN but smaller than WAN. It may comprise the entire network in a city like Mumbai.

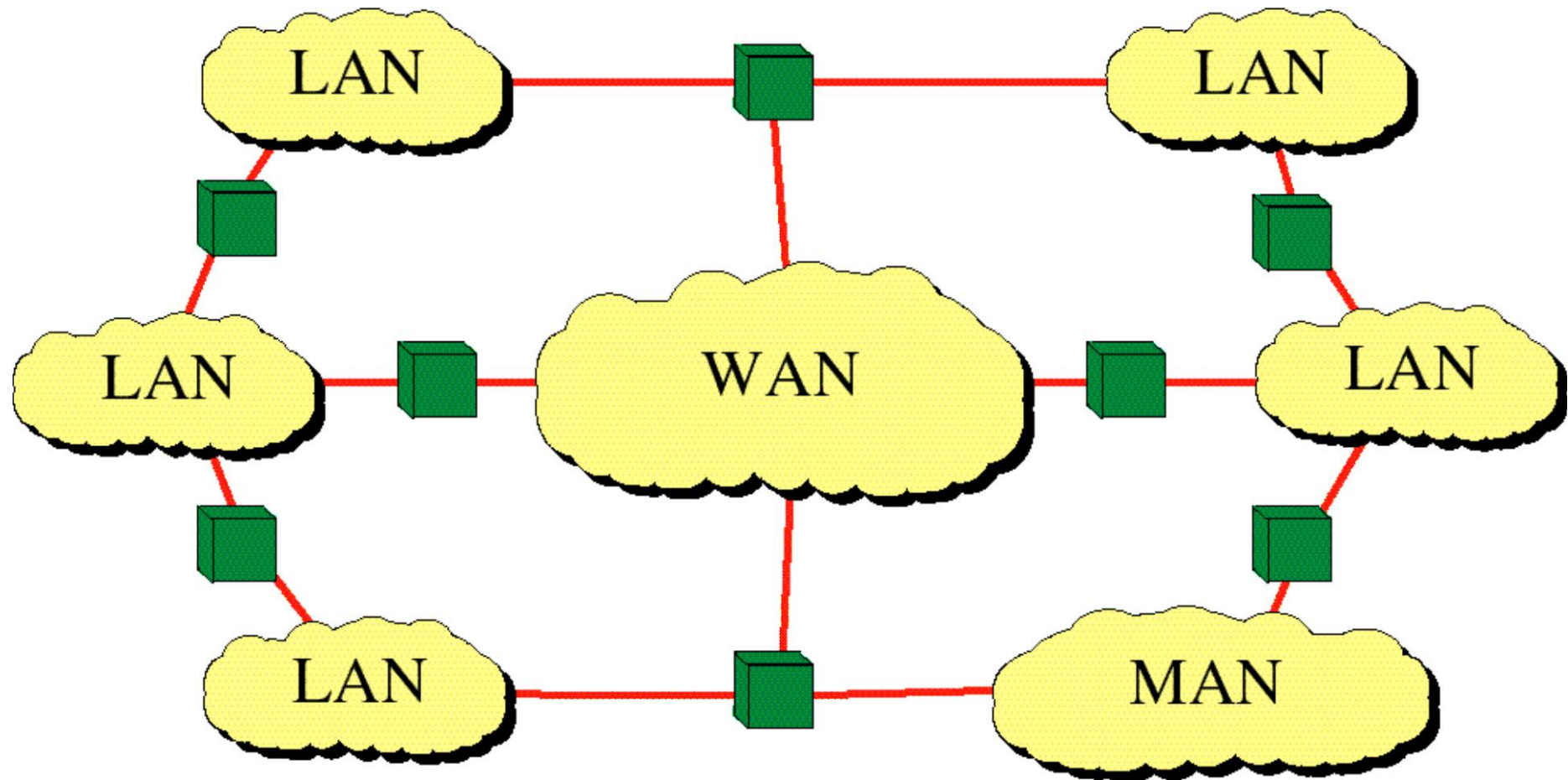


Wide Area Network

Wide Area Network (WAN) is made of all the networks in a (geographically) large area. The network in the entire state of Maharashtra could be a WAN



Internetwork (Internet)

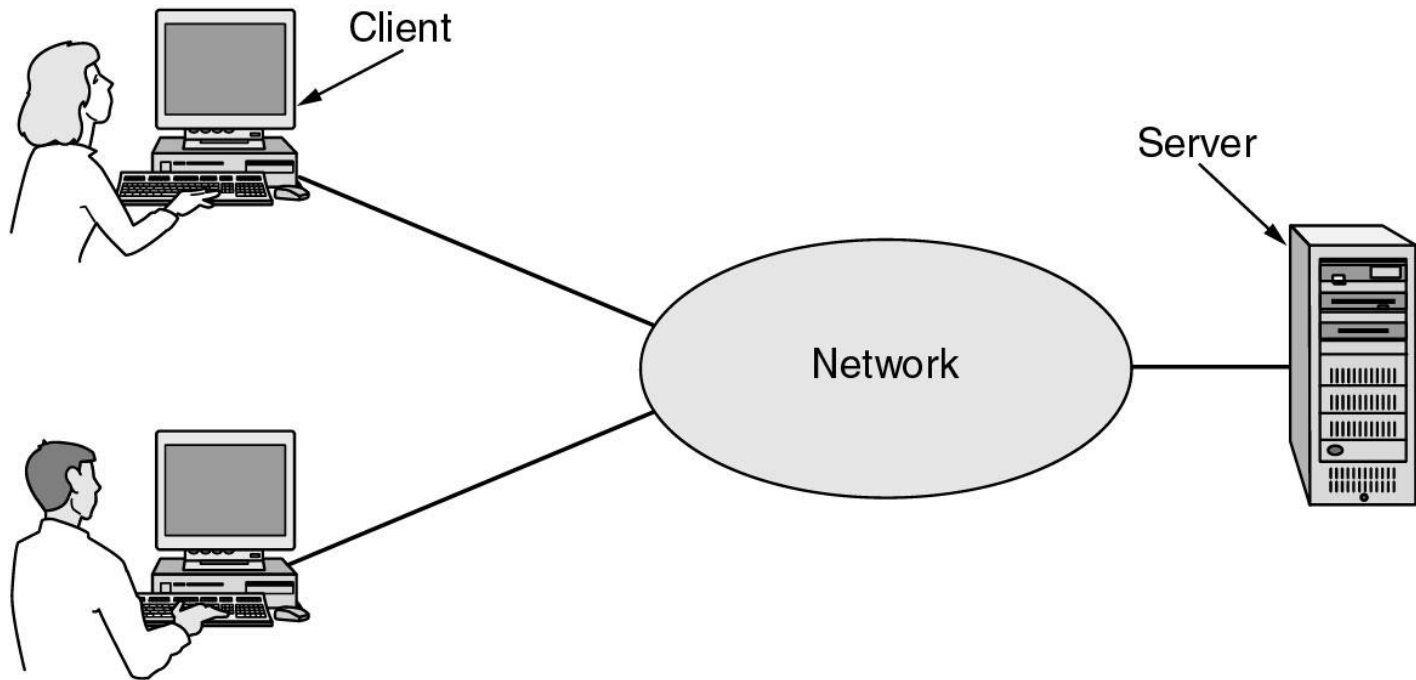


Uses of Computer Networks

- Business Applications
- Home Applications
- Mobile Users
- Social Issues

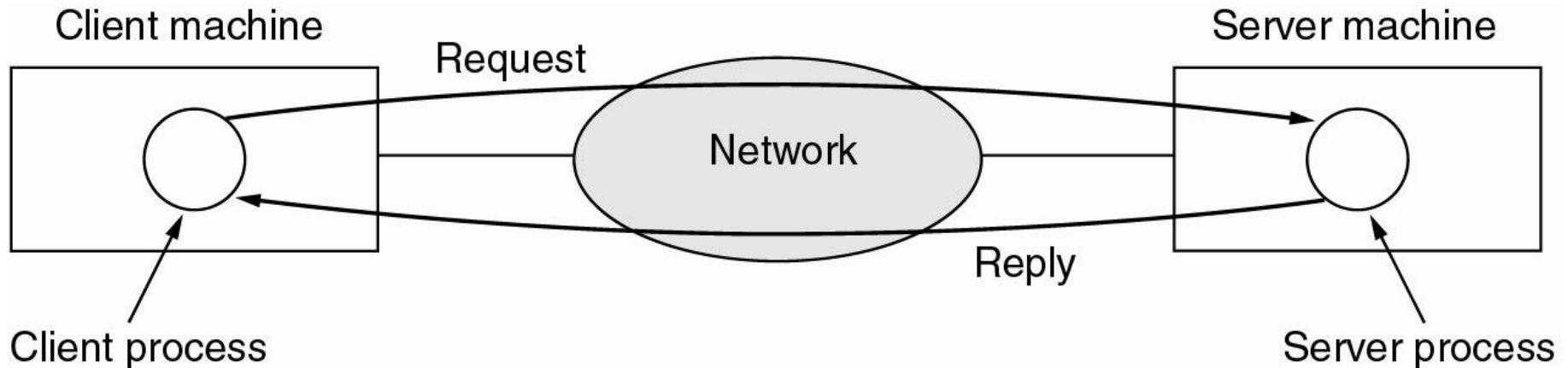
Business Applications of Networks

- A network with two clients and one server.



Business Applications of Networks (2)

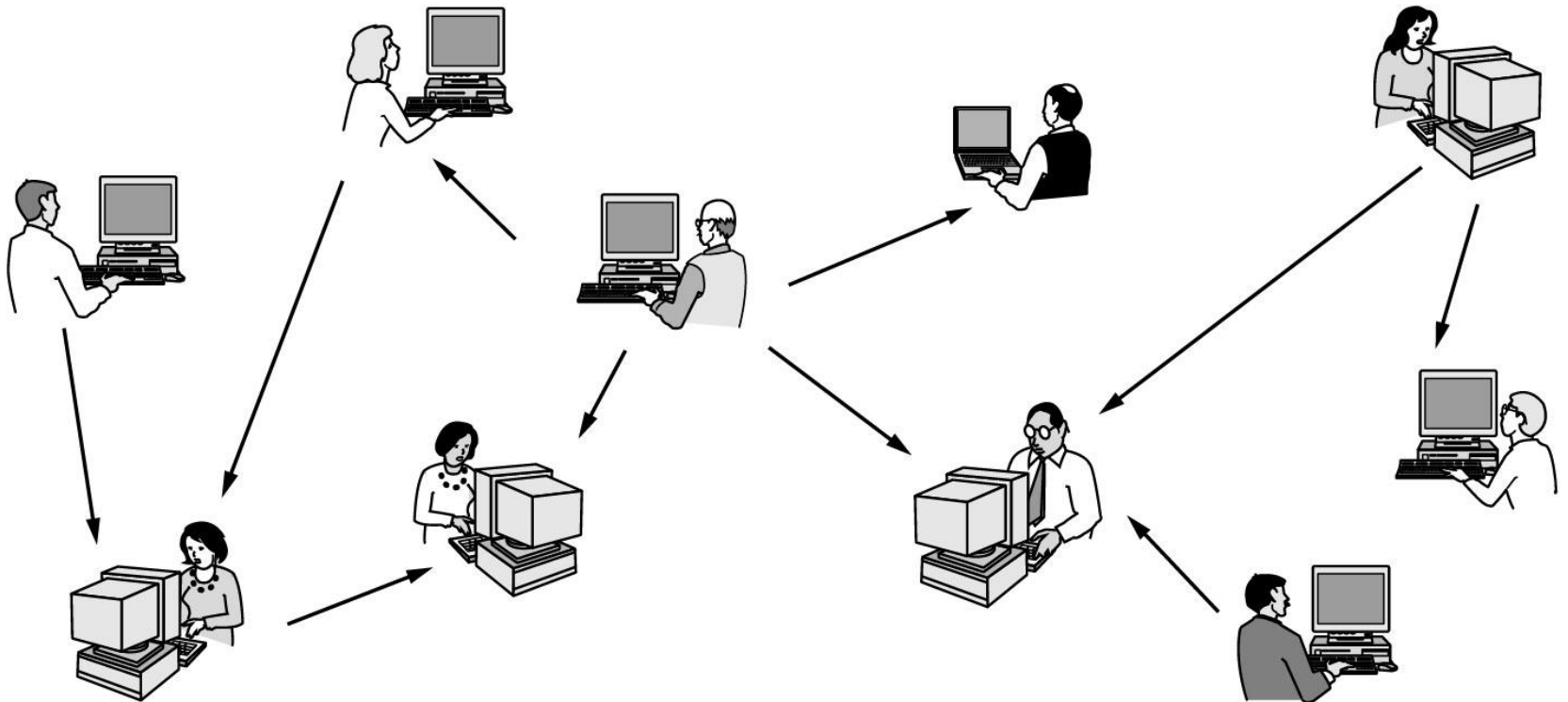
- The client-server model involves requests and replies.



Home Network Applications

- Access to remote information
- Person-to-person communication
- Interactive entertainment
- Electronic commerce

Home Network Applications (2)



Home Network Applications (3)

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books on-line
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products on-line
P2P	Peer-to-peer	File sharing

Mobile Network Users

- Combinations of wireless networks and mobile computing.

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

Network Hardware

- Local Area Networks
- Metropolitan Area Networks
- Wide Area Networks
- Wireless Networks
- Home Networks
- Internetworks

Broadcast Networks

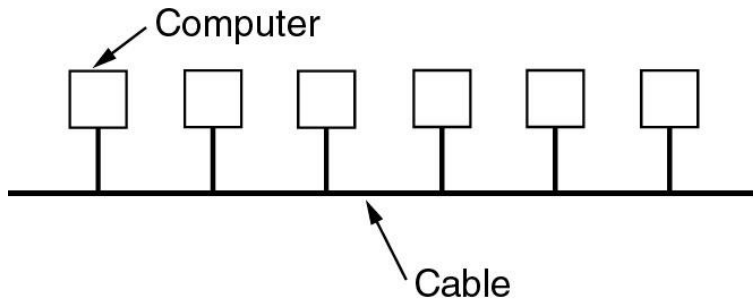
- Types of transmission technology
- Broadcast links
- Point-to-point links

Broadcast Networks (2)

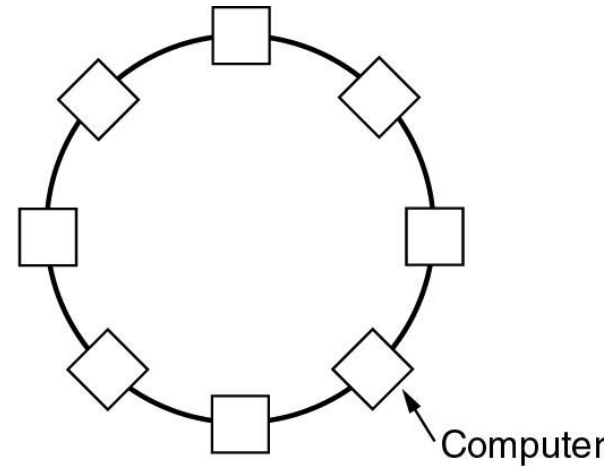
- Classification of interconnected processors by scale.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Local Area Networks



(a)

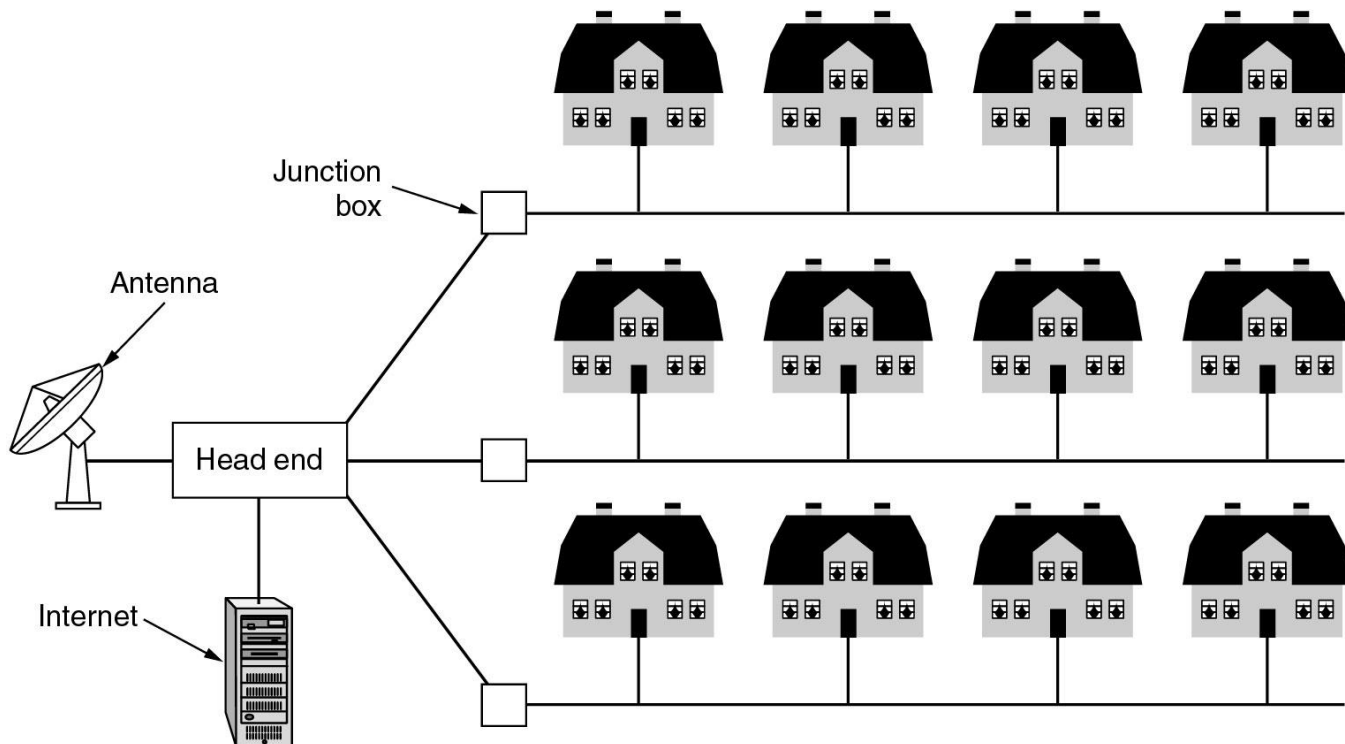


(b)

- Two broadcast networks
- (a) Bus
- (b) Ring

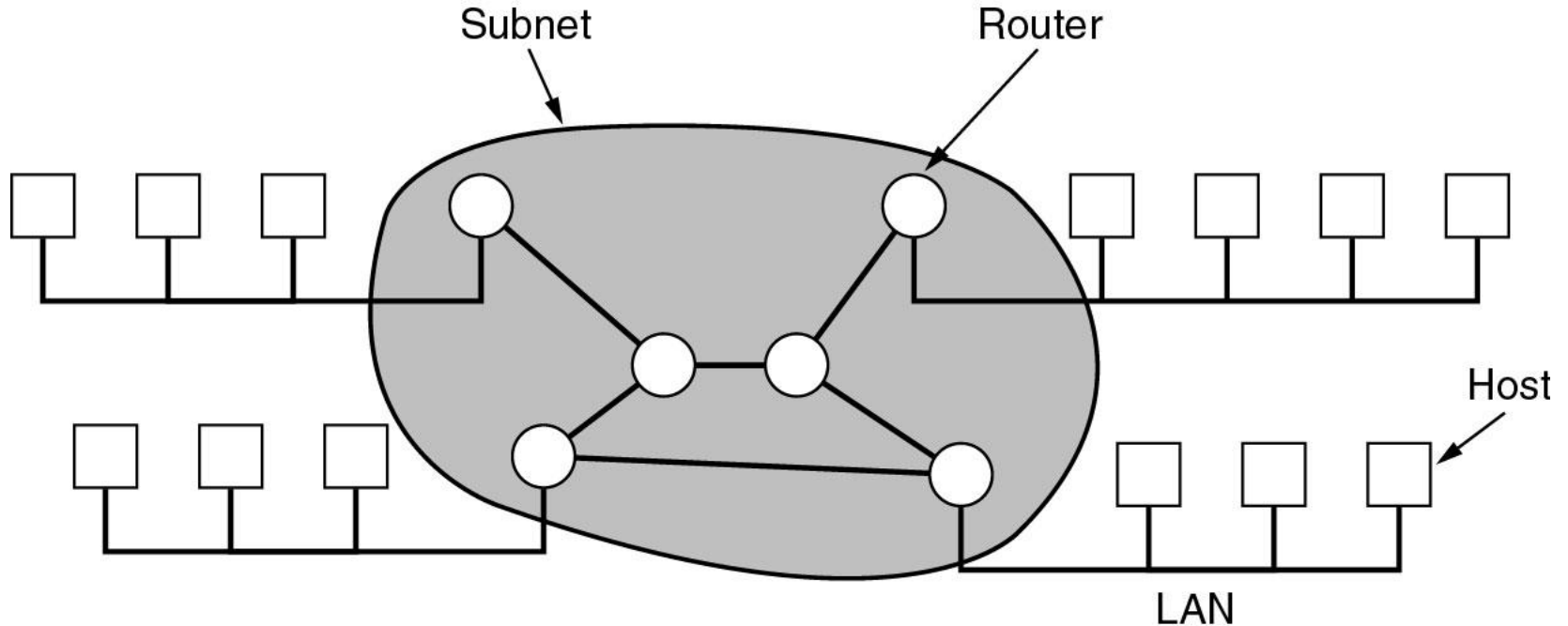
Metropolitan Area Networks

- A metropolitan area network based on cable TV.



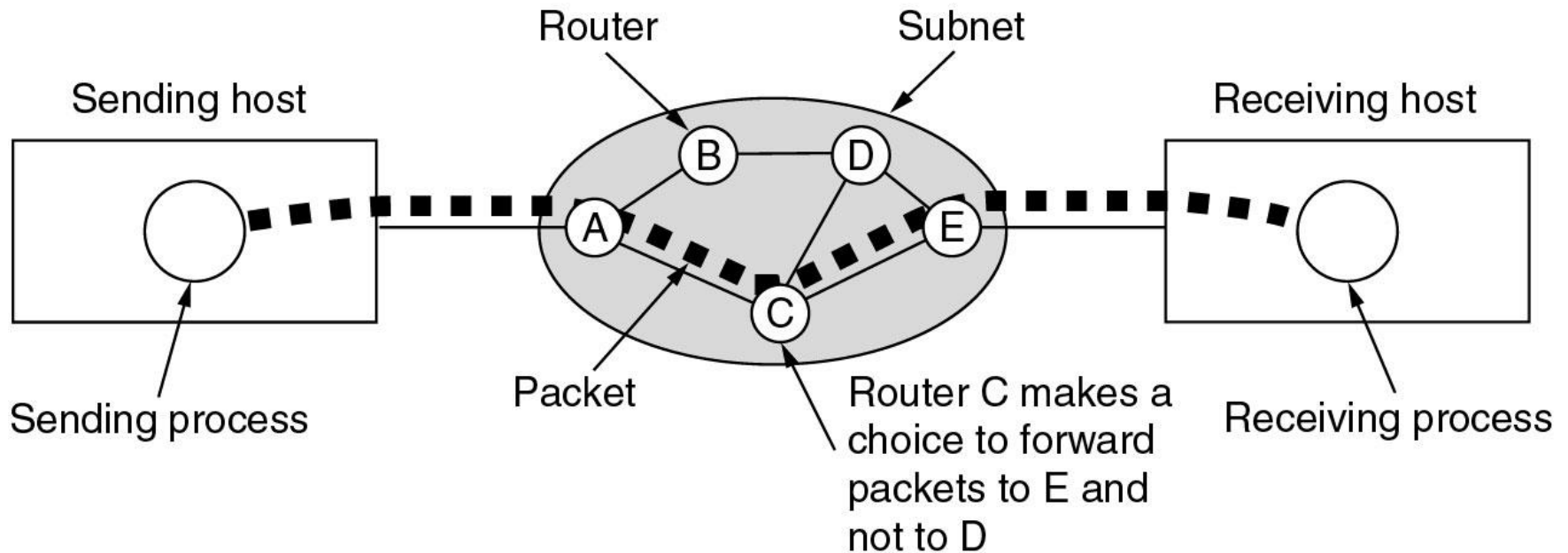
Wide Area Networks

- Relation between hosts on LANs and the subnet.



Wide Area Networks (2)

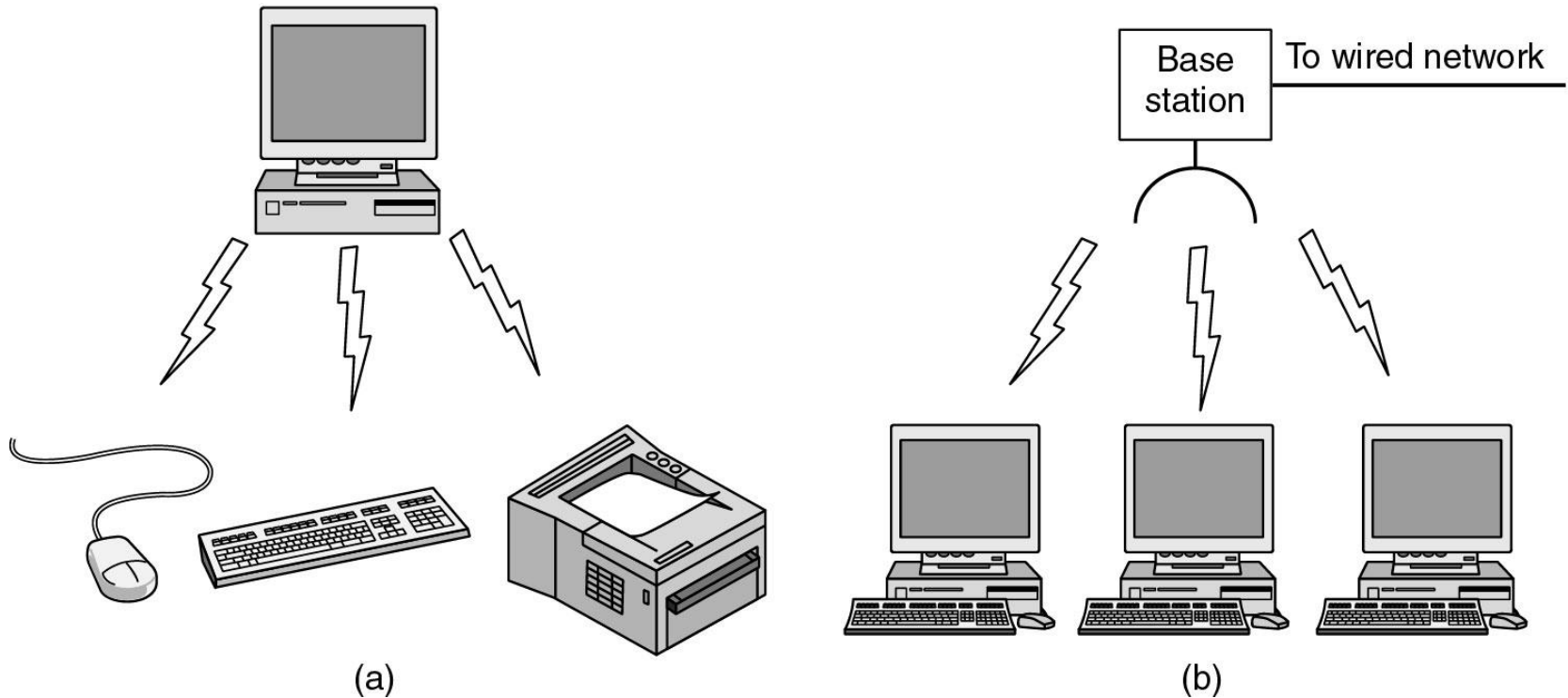
- A stream of packets from sender to receiver.



Wireless Networks

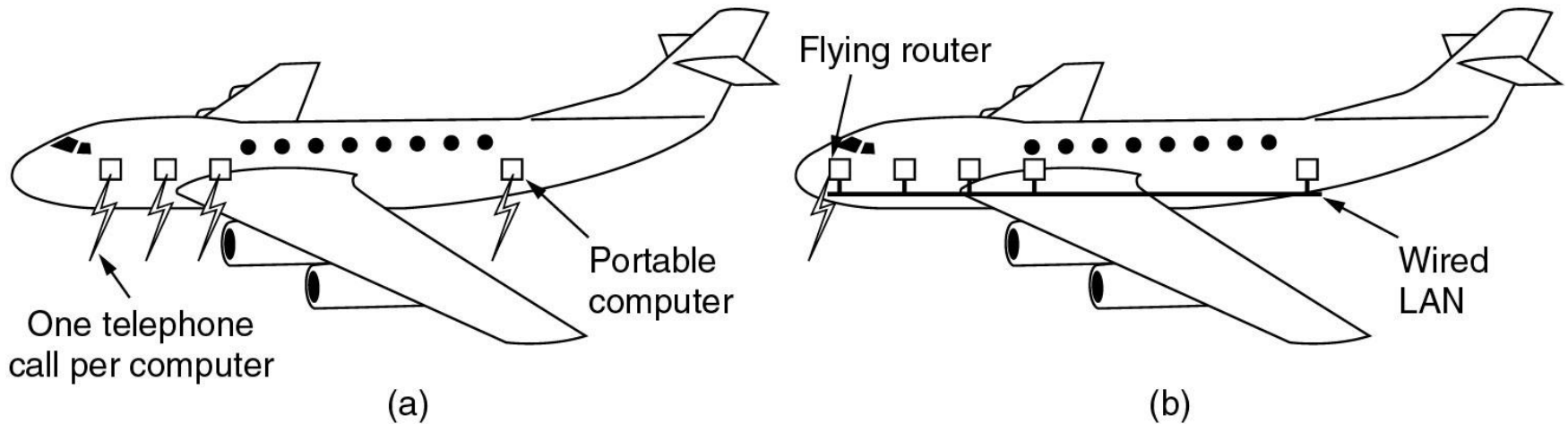
- Categories of wireless networks:
- System interconnection
- Wireless LANs
- Wireless WANs

Wireless Networks (2)



- (a) Bluetooth configuration
- (b) Wireless LAN

Wireless Networks (3)



- (a) Individual mobile computers
- (b) A flying LAN

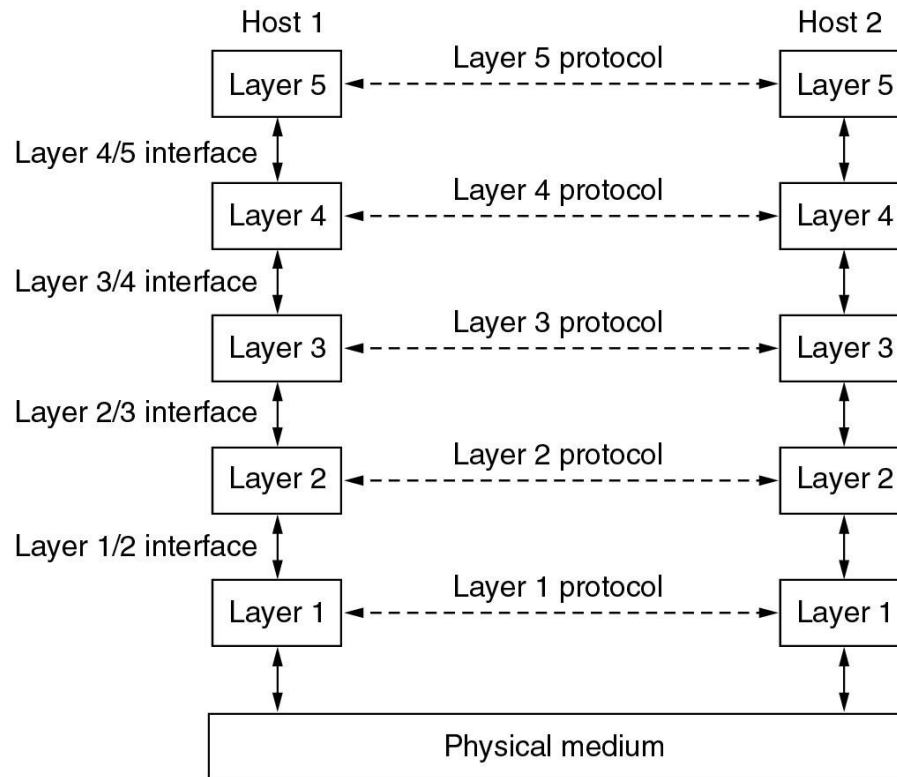
Home Network Categories

- Computers (desktop PC, PDA, shared peripherals)
- Entertainment (TV, DVD, VCR, camera, stereo, MP3)
- Telecomm (telephone, cell phone, intercom, fax)
- Appliances (microwave, fridge, clock, furnace, airco)
- Telemetry (utility meter, burglar alarm, babycam).

Network Software

- Protocol Hierarchies
- Design Issues for the Layers
- Connection-Oriented and Connectionless Services
- Service Primitives
- The Relationship of Services to Protocols

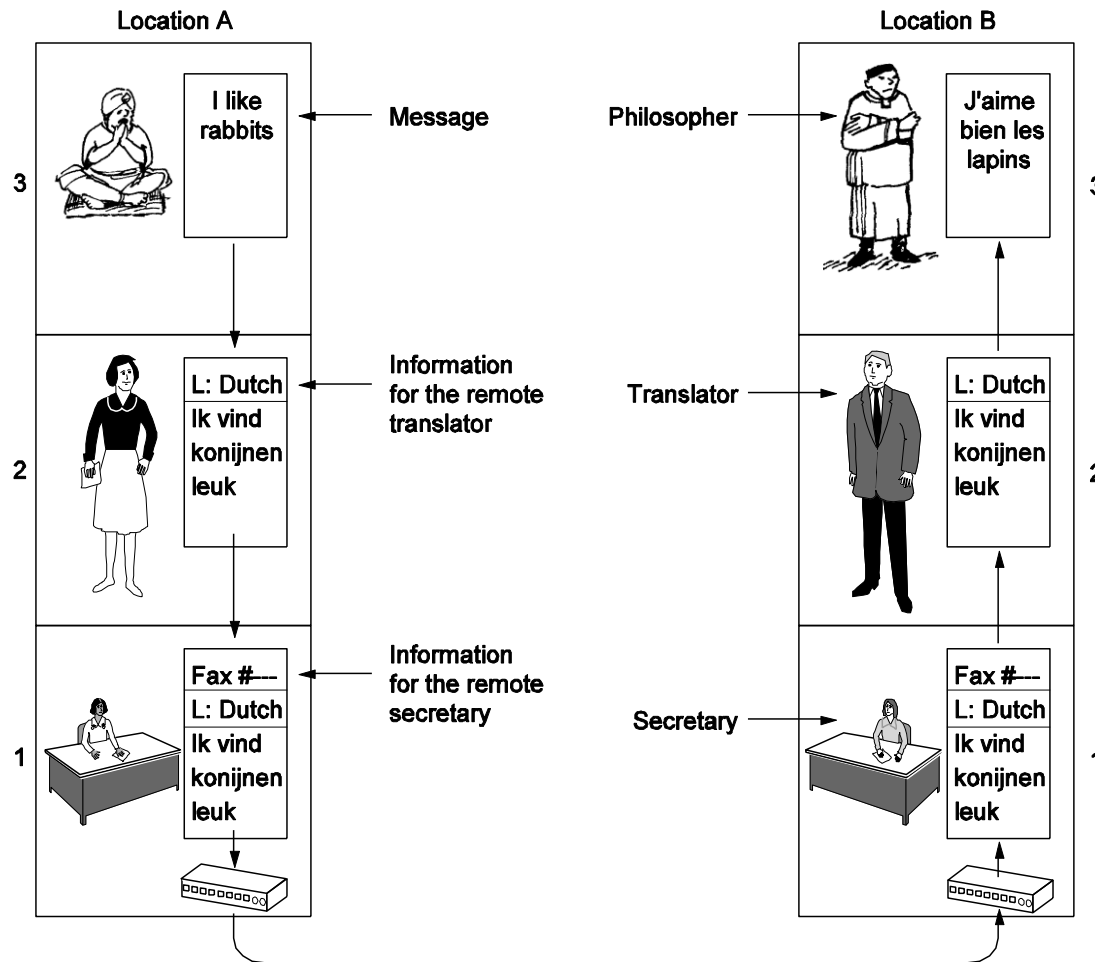
Network Software Protocol Hierarchies



- Layers, protocols, and interfaces.

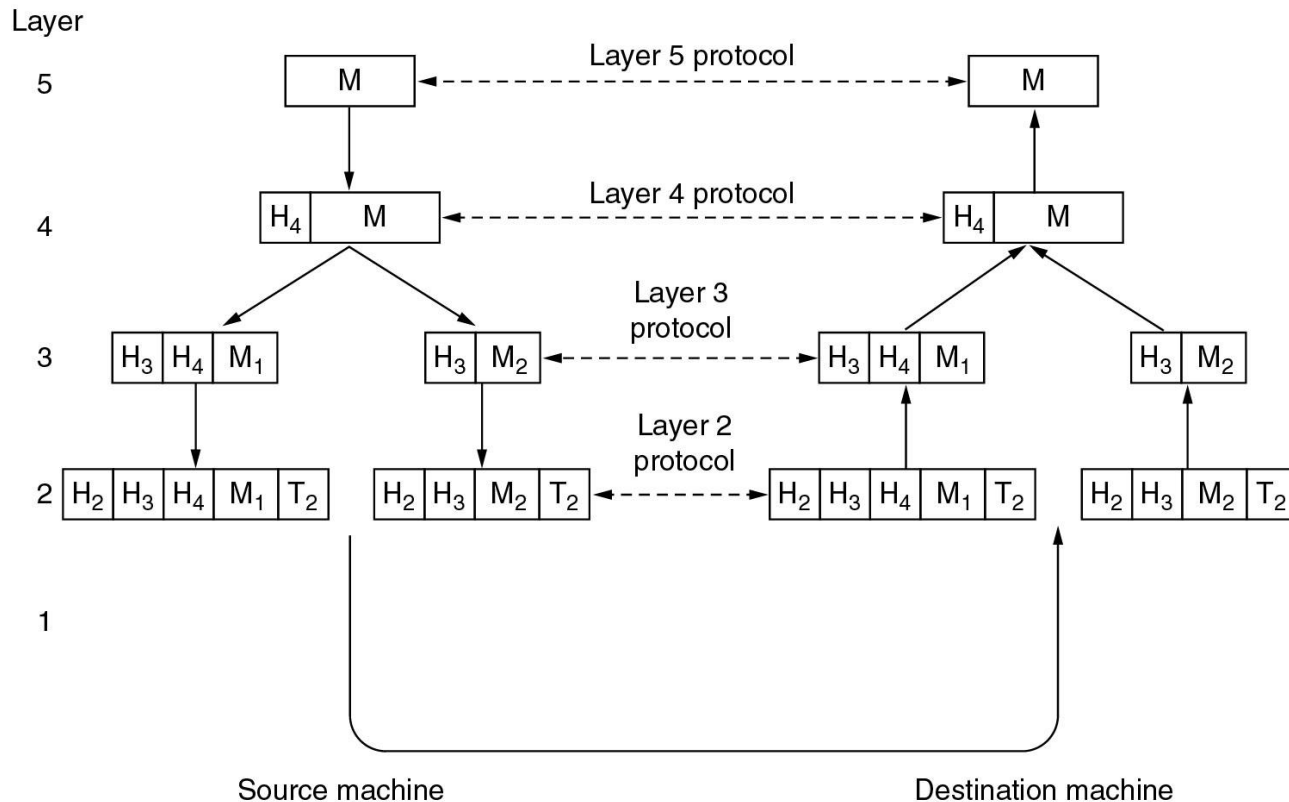
Protocol Hierarchies (2)

- The philosopher-translator-secretary architecture.



Protocol Hierarchies (3)

- Example information flow supporting virtual communication in layer 5.



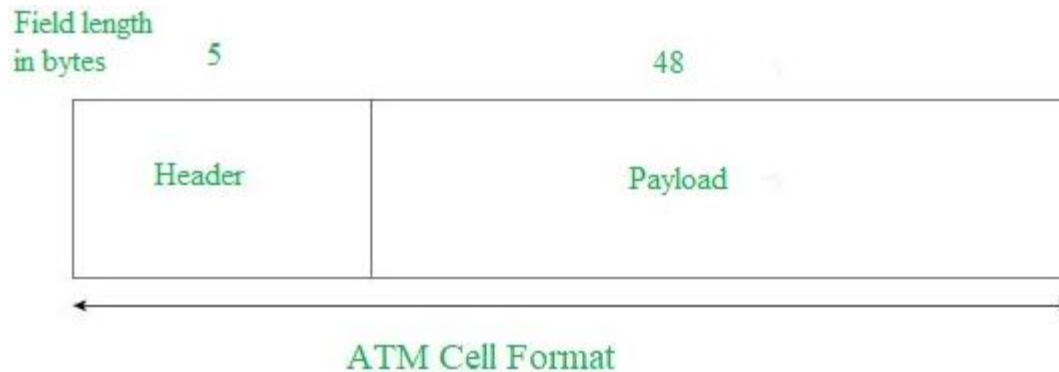
Asynchronous Transfer Mode (ATM):

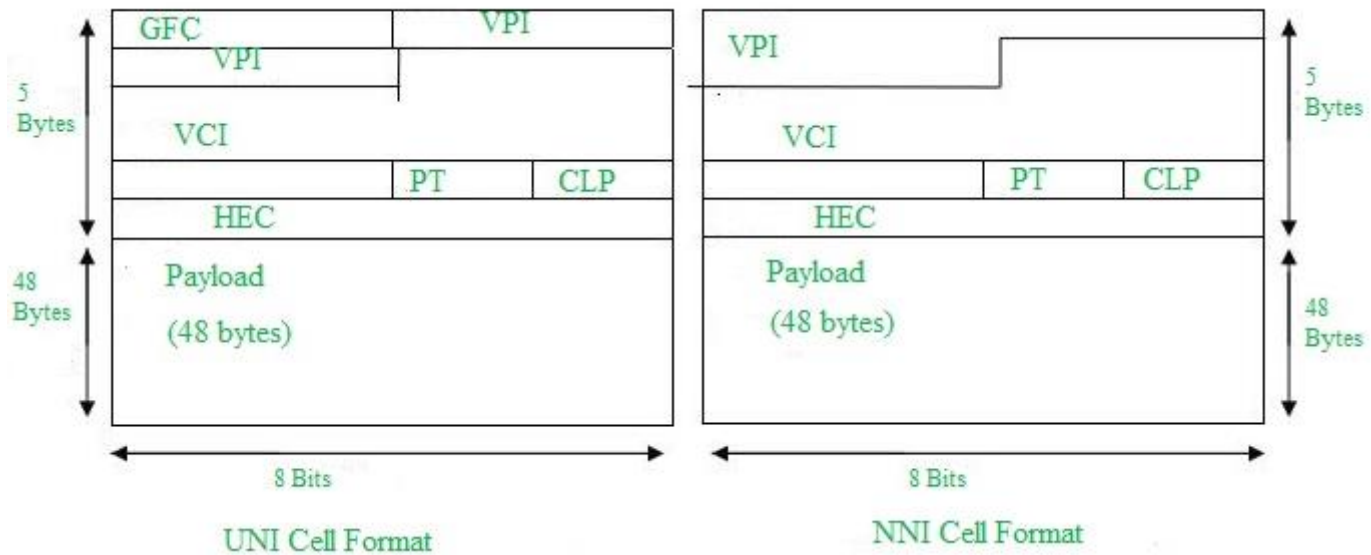
- **Asynchronous Transfer Mode (ATM):**
It is an International Telecommunication Union- Telecommunications Standards Section (ITU-T) efficient for call relay and it transmits all information including multiple service types such as data, video, or voice which is conveyed in small fixed-size packets called cells. Cells are transmitted asynchronously and the network is connection-oriented.
- ATM is a technology that has some event in the development of broadband ISDN in the 1970s and 1980s, which can be considered an evolution of packet switching. *Each cell is 53 bytes long* – 5 bytes header and 48 bytes payload. Making an ATM call requires first sending a message to set up a connection.
- Subsequently, all cells follow the same path to the destination. It can handle both constant rate traffic and variable rate traffic. Thus it can carry multiple types of traffic with end-to-end quality of service. ATM is independent of a transmission medium, they may be sent on a wire or fiber by themselves or they may also be packaged inside the payload of other carrier systems. ATM networks use “Packet” or “cell” Switching with virtual circuits. Its design helps in the implementation of high-performance multimedia networking.

ATM

- As information is transmitted in ATM in the form of fixed-size units called **cells**. As known already each cell is 53 bytes long which consists of a 5 bytes header and 48 bytes payload.

-





1. **UNI Header:** This is used within private networks of ATMs for communication between ATM endpoints and ATM switches. It includes the Generic Flow Control (GFC) field.
2. **NNI Header:** is used for communication between ATM switches, and it does not include the Generic Flow Control(GFC) instead it includes a Virtual Path Identifier (VPI) which occupies the first 12 bits.

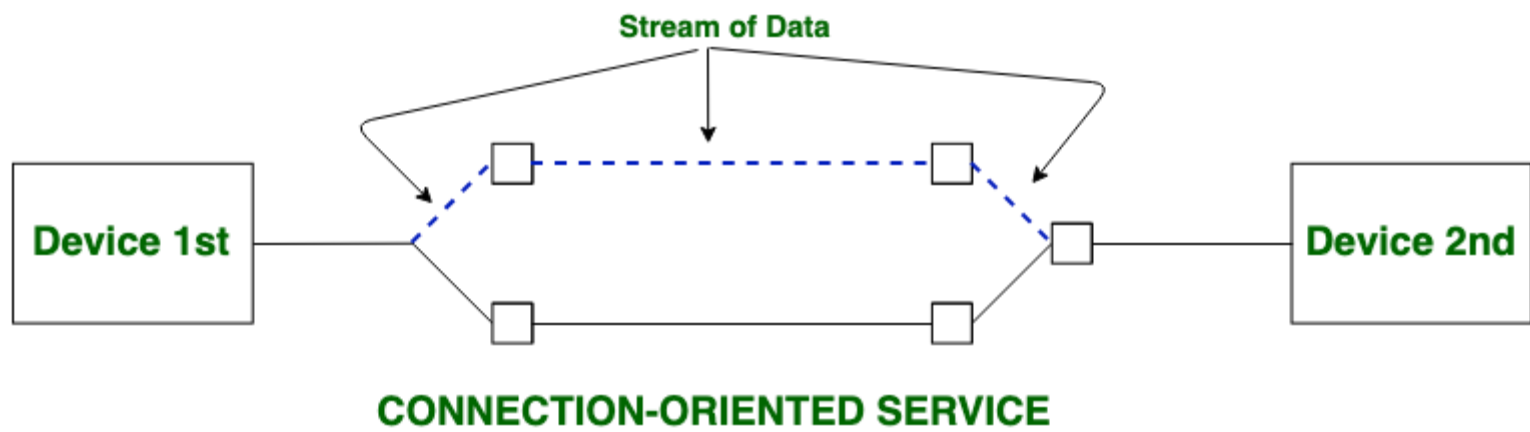
- **Working of ATM:**

ATM standard uses two types of connections. i.e., Virtual path connections (VPCs) which consist of Virtual channel connections (VCCs) bundled together which is a basic unit carrying a single stream of cells from user to user. A virtual path can be created end-to-end across an ATM network, as it does not route the cells to a particular virtual circuit. In case of major failure, all cells belonging to a particular virtual path are routed the same way through the ATM network, thus helping in faster recovery.

- Switches connected to subscribers use both VPIs and VCIs to switch the cells which are Virtual Path and Virtual Connection switches that can have different virtual channel connections between them, serving the purpose of creating a *virtual trunk* between the switches which can be handled as a single entity. Its basic operation is straightforward by looking up the connection value in the local translation table determining the outgoing port of the connection and the new VPI/VCI value of connection on that link.

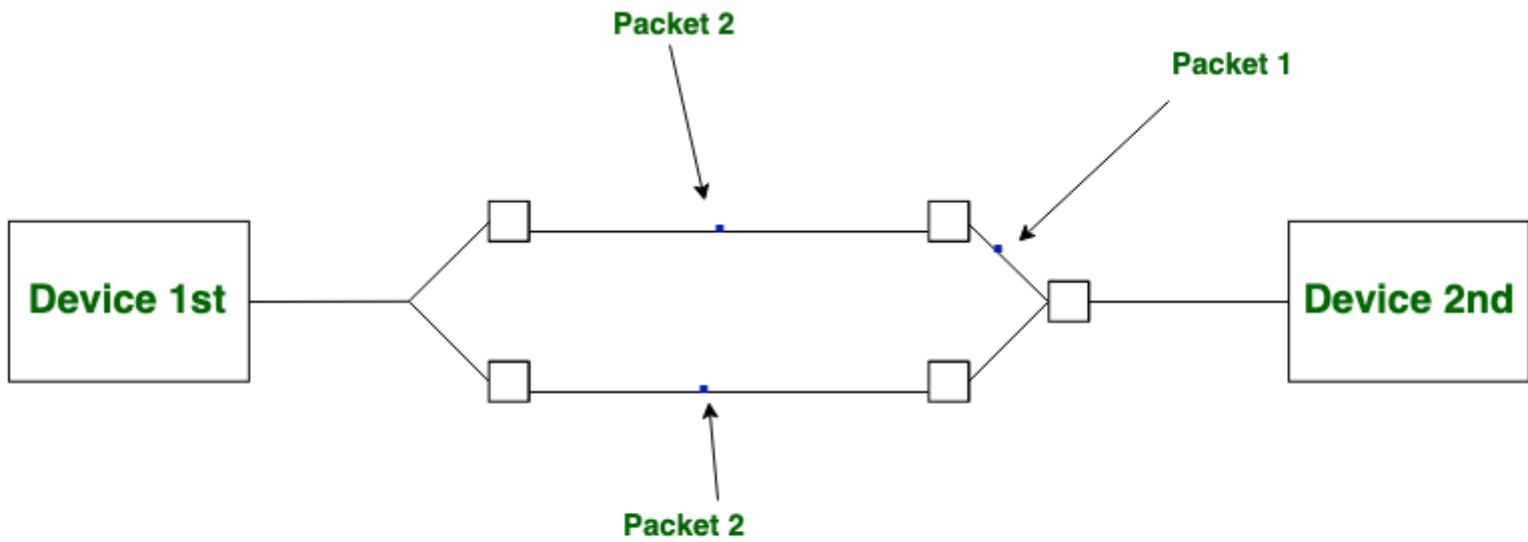
Connection-oriented and Connection-less Services

- **Connection-oriented service**
- Connection-oriented service is related to the telephone system. It includes connection establishment and connection termination. In a connection-oriented service, the Handshake method is used to establish the connection between sender and receiver. Before data transmission starts, connection-oriented services create a dedicated communication channel between the sender and the recipient. As the connection is kept open until all data is successfully transferred, this guarantees dependable data delivery. One example is TCP (Transmission Control Protocol), which ensures error-free and accurate data packet delivery.



- **Connection-Less Service?**

- Connection-less service is related to the postal system. It does not include any connection establishment and connection termination. Connection-less Service does not give a guarantee of reliability. In this, Packets do not follow the same path to reach their destination. Connection-less Services deliver individual data packets without first making a connection. Since each packet is sent separately, delivery, order, and mistake correction cannot be guaranteed. As a result, the service is quicker but less dependable. UDP (User Datagram Protocol) is one example, which is frequently used for streaming where dependability is not as important as speed.



CONNECTIONLESS SERVICE

Difference Between Connection-oriented and Connection-less Services

Connection-oriented Service	Connection-less Service
<u>Connection-oriented</u> service is related to the telephone system.	<u>Connection-less</u> service is related to the postal system.
Connection-oriented service is preferred by long and steady communication.	Connection-less Service is preferred by bursty communication.
Connection-oriented Service is necessary.	Connection-less Service is not compulsory.
Connection-oriented Service is feasible.	Connection-less Service is not feasible.
In connection-oriented Service, Congestion is not possible.	In connection-less Service, Congestion is possible.
Connection-oriented Service gives the guarantee of reliability.	Connection-less Service does not give a guarantee of reliability.
In connection-oriented Service, Packets follow the same route.	In connection-less Service, Packets do not follow the same route.

OSI Reference Model

- The **OSI (Open Systems Interconnection)** Model is a set of rules that explains how different computer systems communicate over a network. OSI Model was developed by the **International Organization for Standardization (ISO)**. The OSI Model consists of 7 layers a
- This layered approach makes it easier for different devices and technologies to work together. OSI Model provides a clear structure for data transmission and managing network issues. The OSI Model is widely used as a reference to understand how network systems function.
- In this article, we will discuss the OSI Model and each layer of the OSI Model in detail. We will also discuss the flow of data in the OSI Model and how the **OSI Model** is different from the **TCP/IP Model**.
- nd each layer has specific functions and responsibilities.

Layers of the OSI Model

There are 7 layers in the OSI Model and each layer has its specific role in handling data. All the layers are mentioned below:

- [Physical Layer](#)
- [Data Link Layer](#)
- [Network Layer](#)
- [Transport Layer](#)
- [Session Layer](#)
- [Presentation Layer](#)
- [Application Layer](#)

OSI Model

- **Layer 1 – Physical Layer**
- The lowest layer of the OSI reference model is the Physical Layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of bits. Physical Layer is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together. Common physical layer devices are Hub, Repeater, Modem, and Cables.



- Functions of the Physical Layer
- Bit Synchronization: The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- Bit Rate Control: The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- Physical Topologies: Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus topology, star topology, or mesh topology.
- Transmission Mode: Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full-duplex.

Layer 2 – Data Link Layer (DLL)

- The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address. Packet in the Data Link layer is referred to as **Frame**. Switches and Bridges are common Data Link Layer devices.

Functions of the Data Link Layer

- **Framing:** Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- **Physical Addressing:** After creating frames, the Data link layer adds physical addresses (**MAC addresses**) of the sender and/or receiver in the header of each frame.
- **Error Control:** The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access Control:** When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.

Layer 3 – Network Layer

- The network layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available. The sender and receiver's IP address are placed in the header by the network layer. Segment in the Network layer is referred to as **Packet**. Network layer is implemented by networking devices such as routers and switches.
- **Functions of the Network Layer**
- **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- **Logical Addressing:** To identify each device inter-network uniquely, the network layer defines an addressing scheme. The sender and receiver's IP addresses are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.

Layer 4 – Transport Layer

- The transport layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as **Segments**. It is responsible for the end-to-end delivery of the complete message. The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found. Protocols used in Transport Layer are [TCP](#), [UDP](#), [NetBIOS](#), [PPTP](#).
- **Functions of the Transport Layer**
- **Segmentation and Reassembly:** This layer accepts the message from the (session) layer, and breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.
- **Service Point Addressing:** To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Layer 5 – Session Layer

Session Layer in the OSI Model is responsible for the establishment of connections, management of connections, terminations of sessions between two devices. It also provides authentication and security. Protocols used in the Session Layer are NetBIOS, PPTP.

Functions of the Session Layer

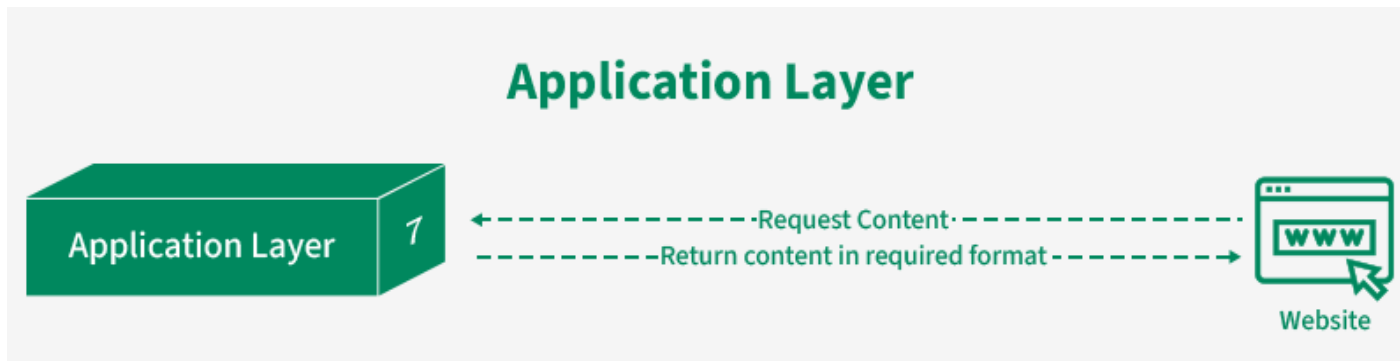
- **Session Establishment, Maintenance, and Termination:** The layer allows the two processes to establish, use, and terminate a connection.
- **Synchronization:** This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- **Dialog Controller:** The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

Layer 6 – Presentation Layer

- The presentation layer is also called the **Translation layer**. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network. Protocols used in the Presentation Layer are [JPEG](#), [MPEG](#), [GIF](#), [TLS/SSL](#), etc.
- **Functions of the Presentation Layer**
- **Translation:** For example, [ASCII to EBCDIC](#).
- **Encryption/ Decryption:** Data encryption translates the data into another form or code. The encrypted data is known as the ciphertext and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- **Compression:** Reduces the number of bits that need to be transmitted on the network.

Layer 7 – Application Layer

At the very top of the OSI Reference Model stack of layers, we find the Application layer which is implemented by the network applications. These applications produce the data to be transferred over the network. This layer also serves as a window for the application services to access the network and for displaying the received information to the user. Protocols used in the Application layer are [SMTP](#), [FTP](#), [DNS](#), etc.

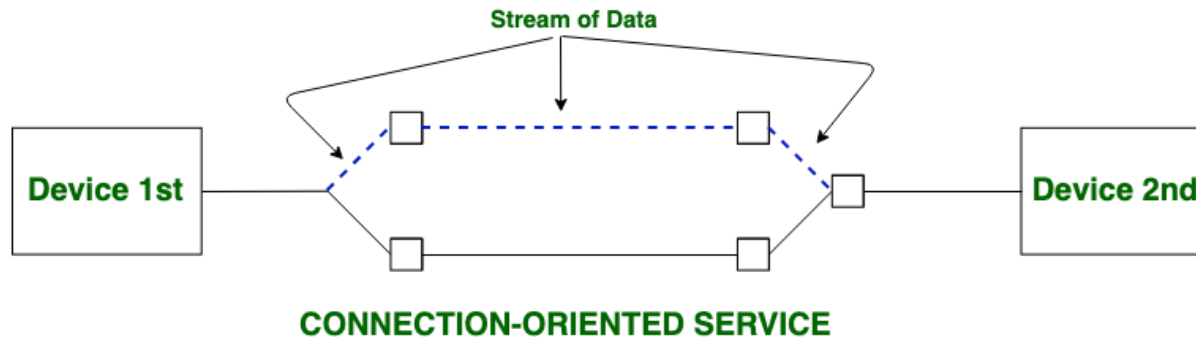


Functions of the Application Layer

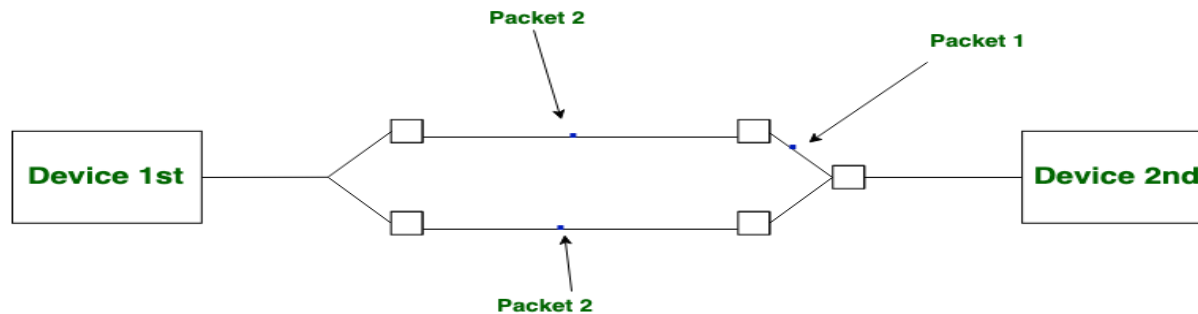
- The main functions of the application layer are given below.
- **Network Virtual Terminal(NVT):** It allows a user to log on to a remote host.
- **File Transfer Access and Management(FTAM):** This application allows a user to access files in a remote host, retrieve files in a remote host, and manage or control files from a remote computer.
- **Mail Services:** Provide email service.
- **Directory Services:** This application provides distributed database sources and access for global information about various objects and services.

Connection Oriented and Connection Less Services

- **Connection-Oriented Service?**
- Connection-oriented service is related to the telephone system. It includes connection establishment and connection termination. In a connection-oriented service, the Handshake method is used to establish the connection between sender and receiver. Before data transmission starts, connection-oriented services create a dedicated communication channel between the sender and the recipient.



- **Connection-Less Service?**
- Connection-less service is related to the postal system. It does not include any connection establishment and connection termination. Connection-less Service does not give a guarantee of reliability. In this, Packets do not follow the same path to reach their destination. Connection-less Services deliver individual data packets without first making a connection. Since each packet is sent separately, delivery, order, and mistake correction cannot be guaranteed. As a result, the service is quicker but less dependable. UDP (User Datagram Protocol) is one example, which is frequently used for streaming where dependability is not as important as speed.



CONNECTIONLESS SERVICE

Connection-oriented Service	Connection-less Service
<u>Connection-oriented</u> service is related to the telephone system.	<u>Connection-less</u> service is related to the postal system.
Connection-oriented service is preferred by long and steady communication.	Connection-less Service is preferred by bursty communication.
Connection-oriented Service is necessary.	Connection-less Service is not compulsory.
Connection-oriented Service is feasible.	Connection-less Service is not feasible.
In connection-oriented Service, Congestion is not possible.	In connection-less Service, Congestion is possible.
Connection-oriented Service gives the guarantee of reliability.	Connection-less Service does not give a guarantee of reliability.
In connection-oriented Service, Packets follow the same route.	In connection-less Service, Packets do not follow the same route.
Connection-oriented services require a bandwidth of a high range.	Connection-less Service requires a bandwidth of low range.
Ex: <u>TCP (Transmission Control Protocol)</u>	Ex: <u>UDP (User Datagram Protocol)</u>
Connection-oriented requires authentication.	Connection-less Service does not require authentication.

Types of Transmission Media in Computer Networks

Transmission media refers to the physical medium through which data is transmitted from one device to another within a network. These medium can be wired or wireless. The choice of medium depends on factors like distance, speed, and interference. In this article, we will discuss the transmission media. In this article we will see types of transmission media in detail.

- **1. Guided Media**
- Guided Media is also referred to as Wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links.
Features:
 - High Speed
 - Secure
 - Used for comparatively shorter distances

Twisted Pair Cable

- It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media. Twisted Pair is of two types:
- **Unshielded Twisted Pair (UTP):** UTP consists of two insulated copper wires twisted around one another. This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.



Unshielded Twisted Pair

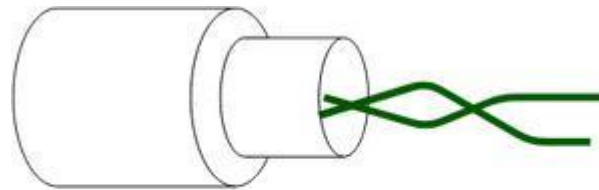
- **Advantages of Unshielded Twisted Pair**

- Least expensive
- Easy to install
- High-speed capacity

- **Disadvantages of Unshielded Twisted Pair**

- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation
-

- **Shielded Twisted Pair (STP):** Shielded Twisted Pair (STP) cable consists of a special jacket (a copper braid covering or a foil shield) to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.



Shielded Twisted Pair

- **Advantages of Shielded Twisted Pair**
- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparatively faster
- **Disadvantages of Shielded Twisted Pair**
- Comparatively difficult to install and manufacture
- More expensive
- Bulky

Coaxial Cable

- Coaxial cable has an outer plastic covering containing an insulation layer made of PVC or Teflon and 2 parallel conductors each having a separate insulated protection cover. The coaxial cable transmits information in two modes: Baseband mode(dedicated cable bandwidth) and Broadband mode(cable bandwidth is split into separate ranges). Cable TVs and analog television networks widely use Coaxial cables.

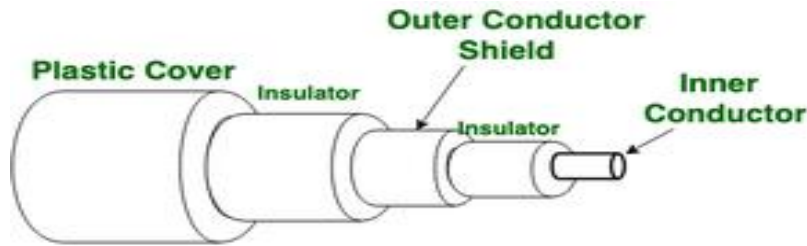


Figure of Coaxial Cable

Advantages of Coaxial Cable

- Coaxial cables has high bandwidth.
- It is easy to install.
- Coaxial cables are more reliable and durable.
- Less affected by noise or cross-talk or electromagnetic inference.
- Coaxial cables support multiple channels

Disadvantages of Coaxial Cable

- Coaxial cables are expensive.
- The coaxial cable must be grounded in order to prevent any crosstalk.
- As a Coaxial cable has multiple layers it is very bulky.
- There is a chance of breaking the coaxial cable and attaching a “t-joint” by hackers, this compromises the security of the data.

Optical Fiber Cable

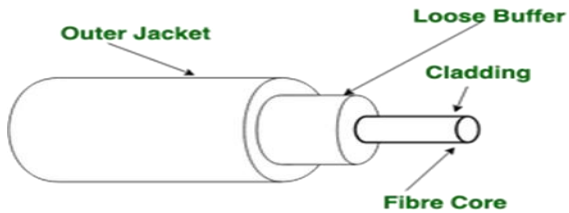


Figure of Optical Fibre Cable

Optical Fibre Cable uses the concept of refraction of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the coating. It is used for the transmission of large volumes of data. The cable can be unidirectional or bidirectional. The WDM (Wavelength Division Multiplexer) supports two modes, namely unidirectional and bidirectional mode.

- **Advantages of Optical Fibre Cable**

- Increased capacity and bandwidth
- Lightweight
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials

Disadvantages of Optical Fibre Cable

- Difficult to install and maintain
- High cost

Applications of Optical Fibre Cable

- **Medical Purpose:** Used in several types of medical instruments.
- **Defence Purpose:** Used in transmission of data in aerospace.
- **For Communication:** This is largely used in formation of internet cables.
- **Industrial Purpose:** Used for lighting purposes and safety measures in designing the interior and exterior of automobiles.